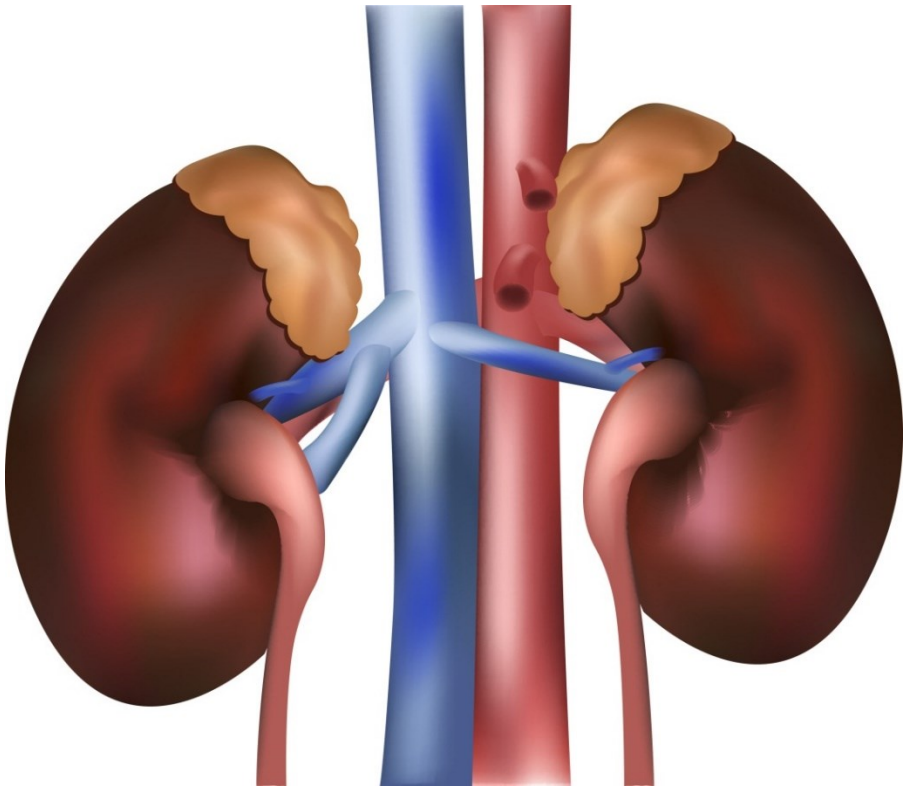
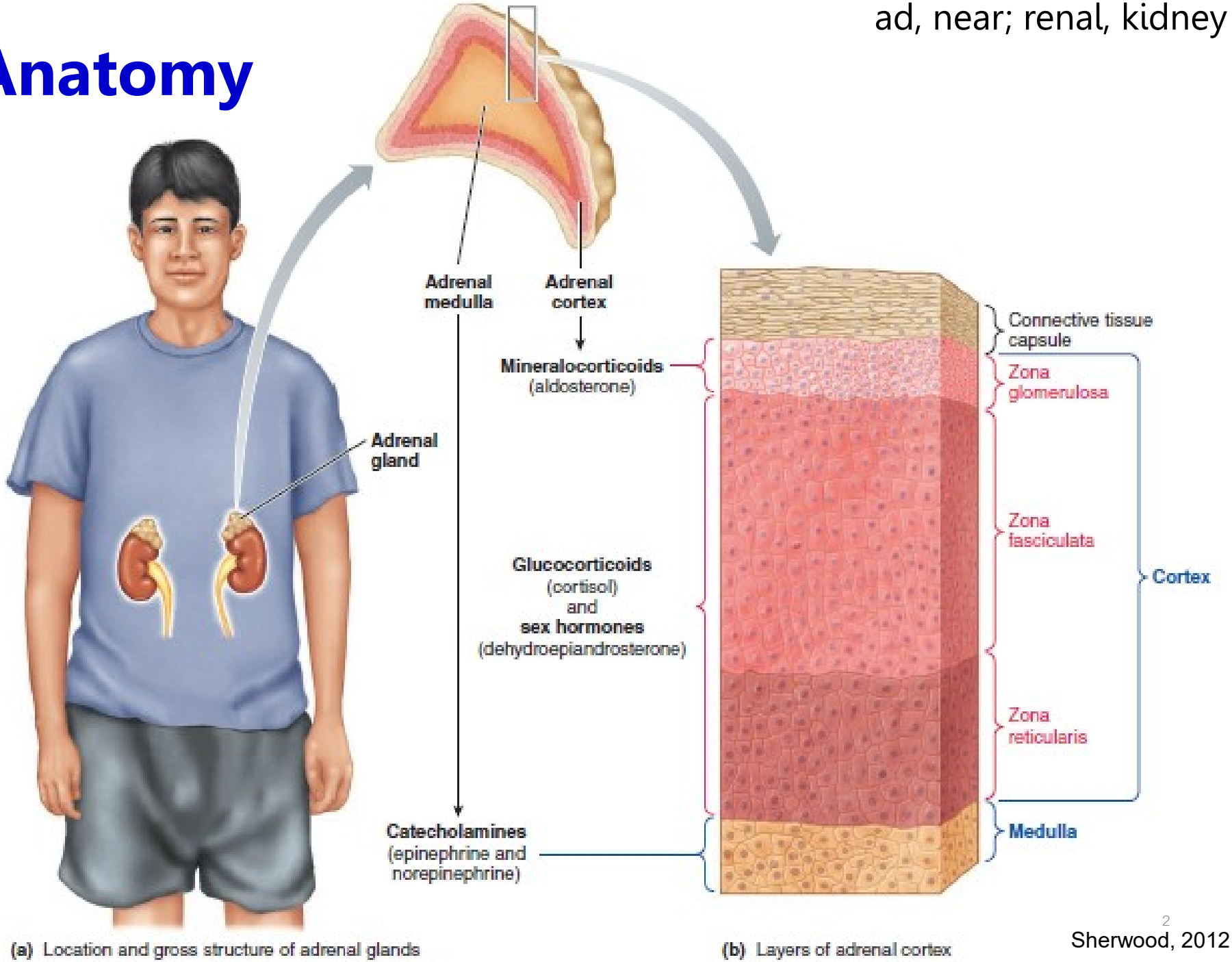
Adrenal gland



Assoc. Prof. Pissared Muangnil, Ph.D.
Faculty of Science, PSU

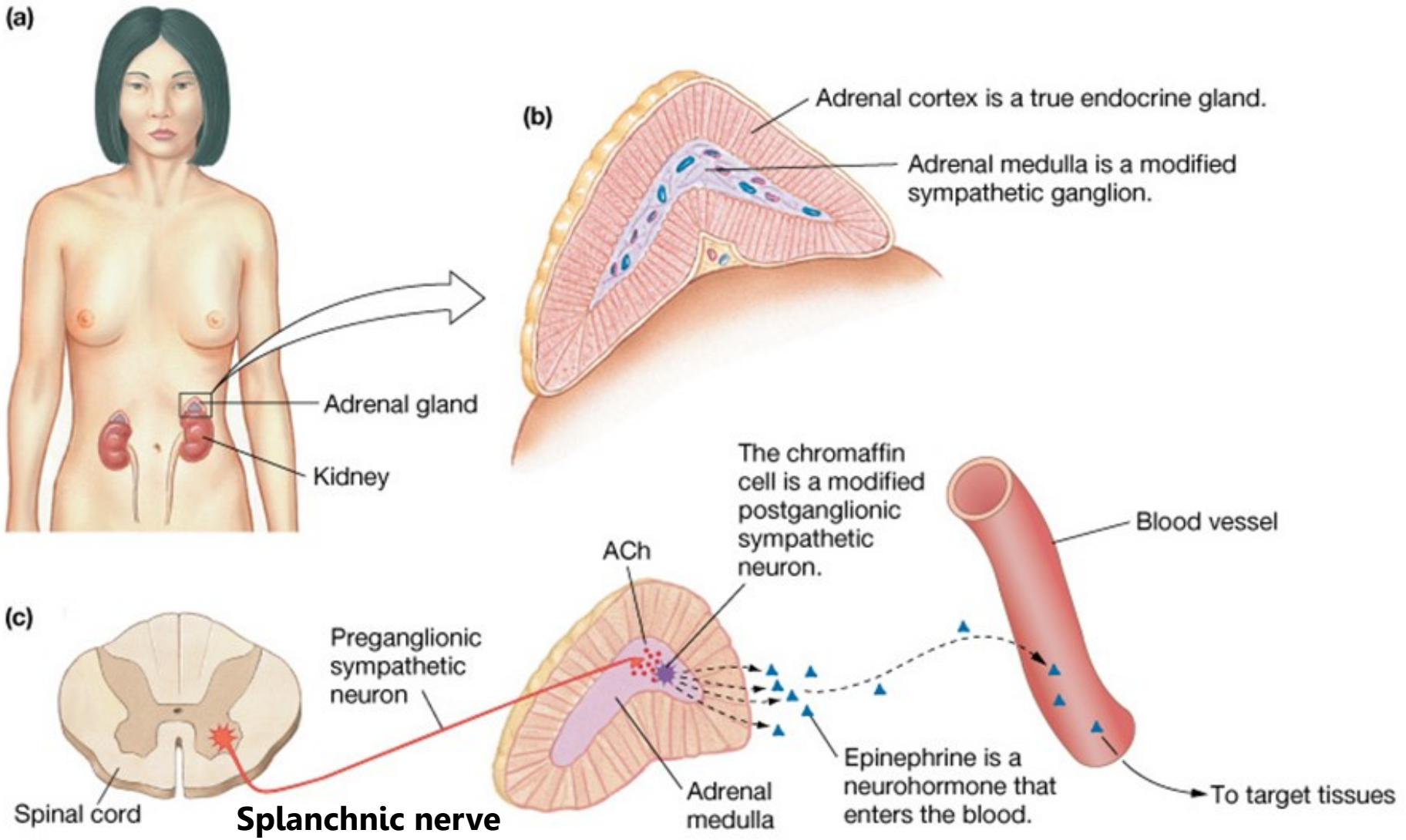
ad, near; renal, kidney

Anatomy



Sherwood, 2012

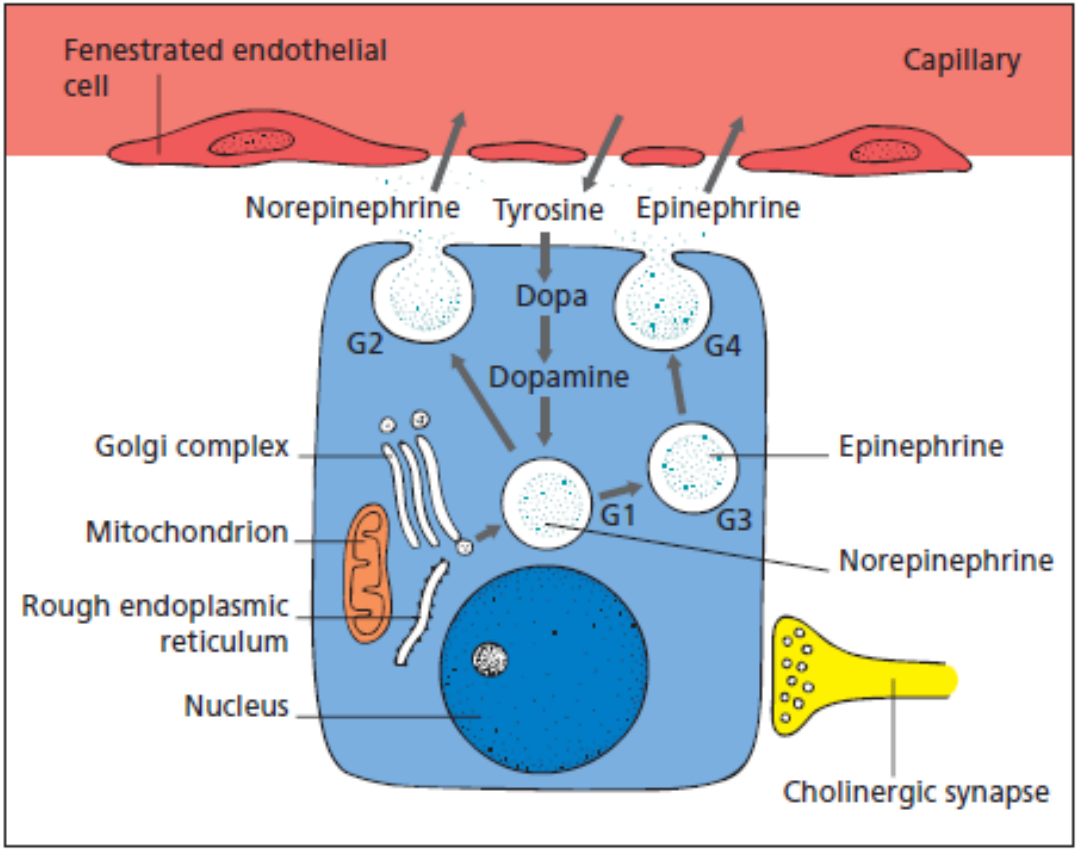
Adrenal medulla



Neurotransmitter *vs.* Neurohormone

- **80% → epinephrine**
  Entirely from the adrenal medulla
- **20% → norepinephrine**
  30% from the medulla
  70% from postganglionic sympathetic nerve terminals
- **Not essential for life**

Synthesis of catecholamines



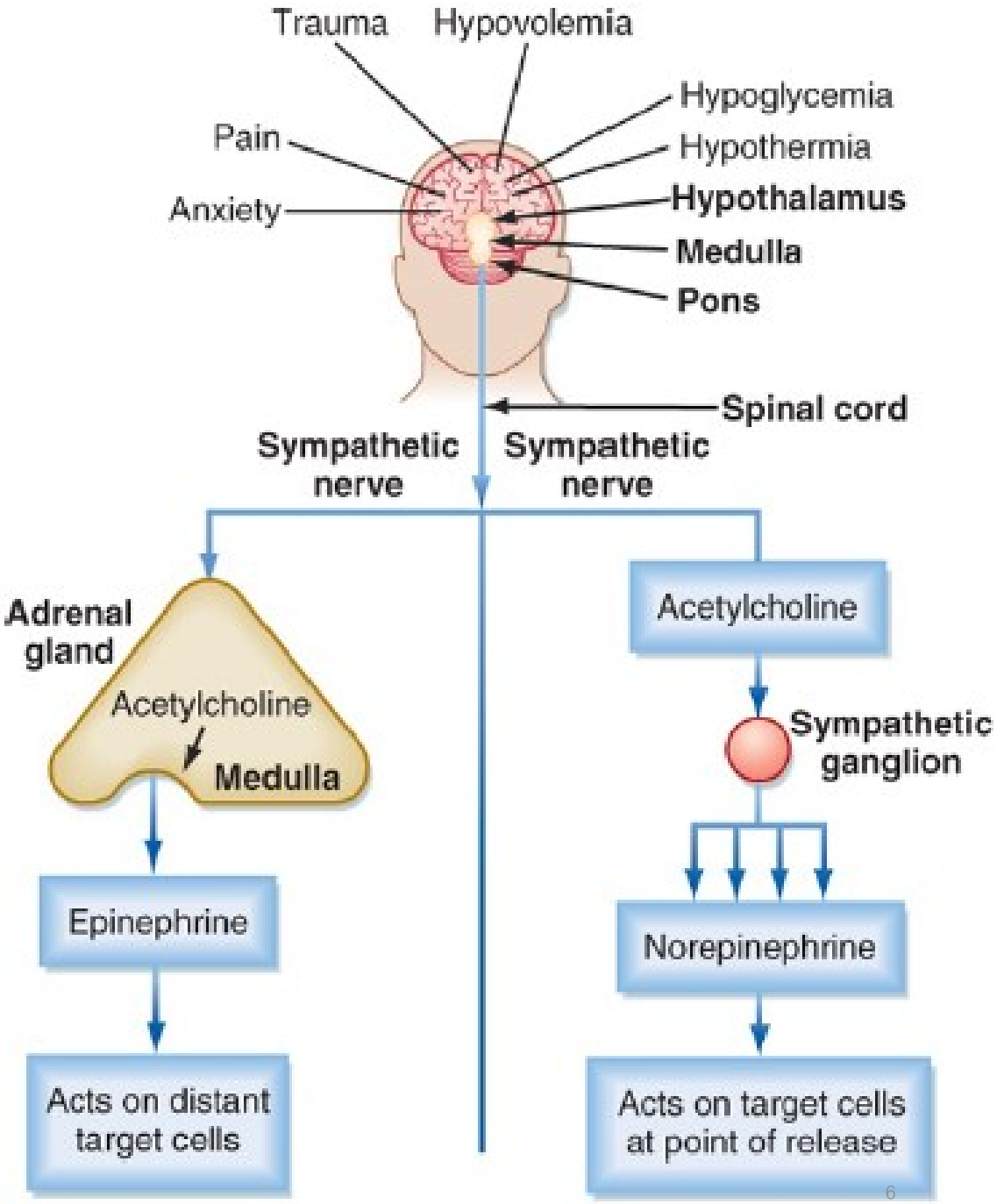
- **Chromaffin cell** (pheochromocyte)
- Storage granules (G), including the **chromogranins**, are synthesized on RER and packaged in the Golgi complex.
- Dopamine enters G1 and is converted to NE, which can then be released exocytotically (G2).
- NE can be further converted to E (G3) & released by exocytosis (G4).
- A cell has a cholinergic synapse.

Holt and Hanley, 2012

chromogranins serve as clinical biomarkers of endocrine tumours.

Stimuli enhance catecholamines secretion.

- Stress
  - Exercise
  - Hypoglycemia
  - Hemorrhagic hypovolemia
- Ach → ⇧ activity of the rate-limiting enzyme tyrosine hydroxylase & dopamine beta-hydroxylase, ⇧ exocytosis of chromaffin granules

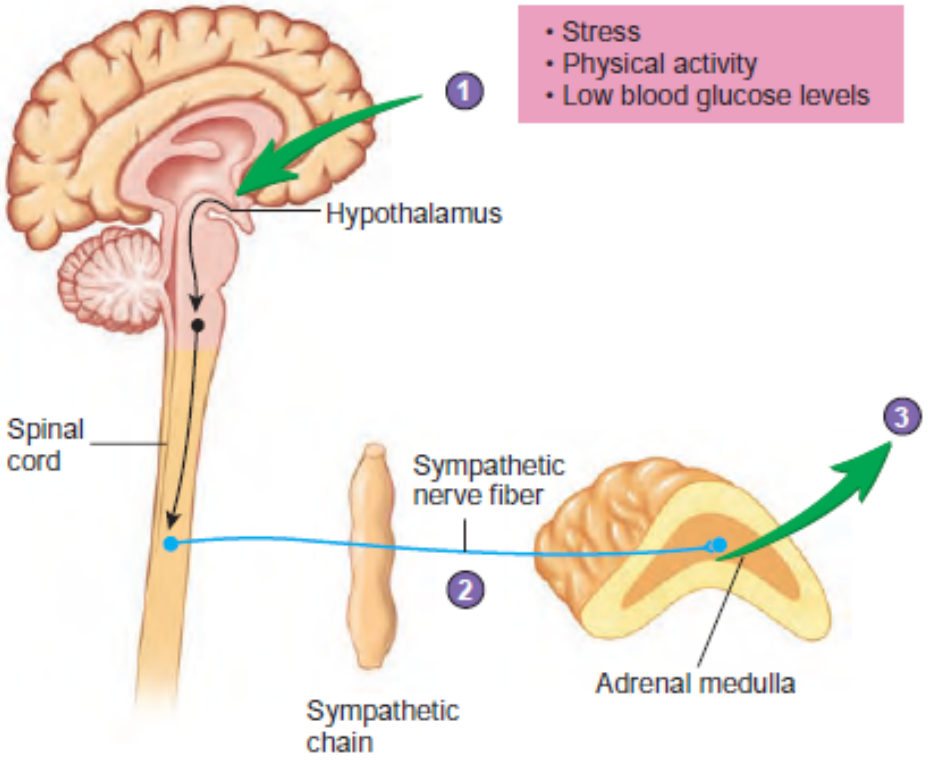


Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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1. Stress, physical activity, and low blood glucose levels act as stimuli to the hypothalamus, resulting in increased sympathetic nervous system activity.

2. An increased frequency of action potentials conducted through the sympathetic division of the autonomic nervous system stimulates the adrenal medulla to secrete epinephrine and some norepinephrine into the circulatory system.

3. Epinephrine and norepinephrine act on their target tissues to produce responses.



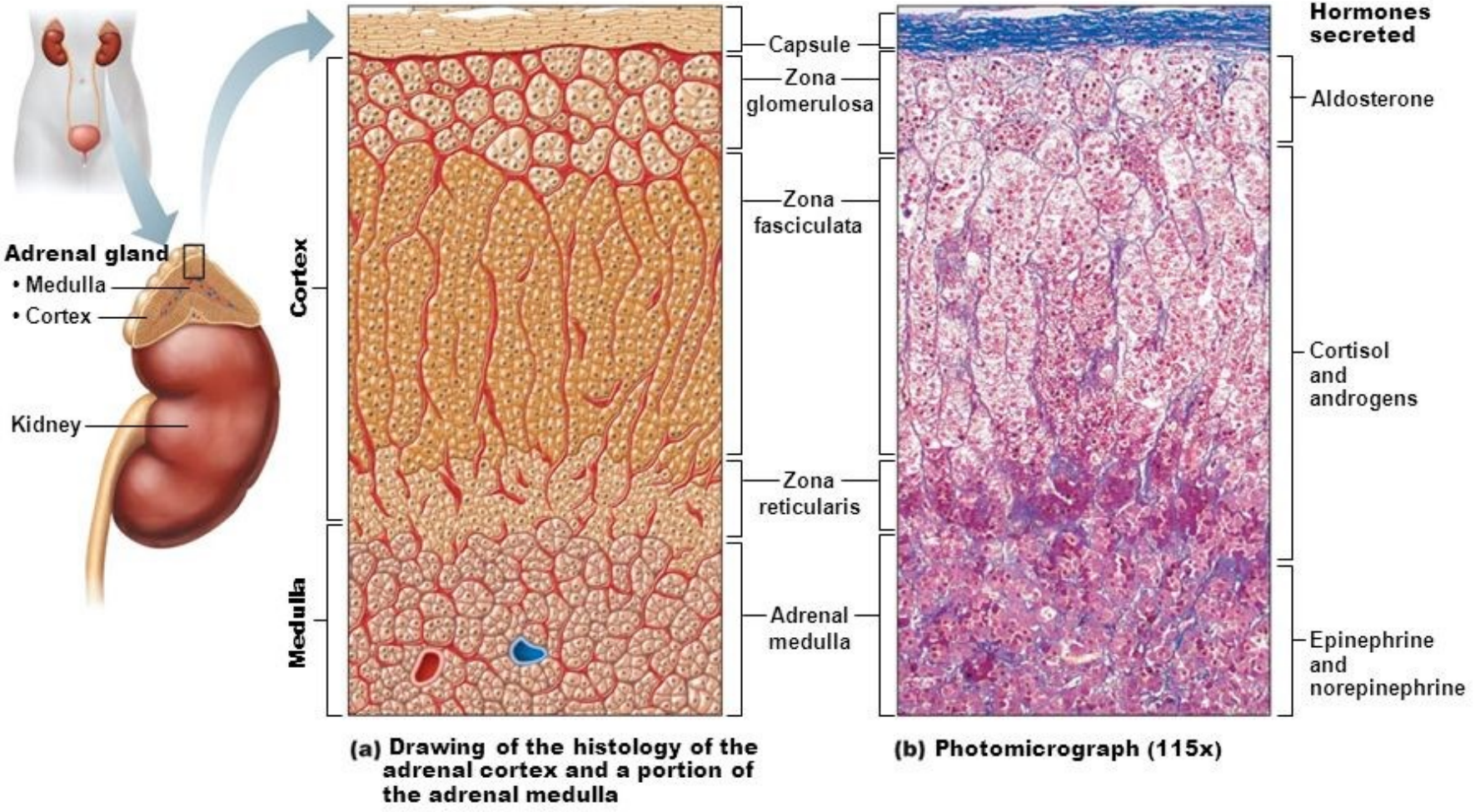
- Stress
- Physical activity
- Low blood glucose levels

- Increases the release of glucose from the liver into the blood
- Increases the release of fatty acids from adipose tissue into the blood
- Increases heart rate
- Decreases blood flow through blood vessels of internal organs
- Increases blood flow through blood vessels of skeletal muscle and the heart
- Increases blood pressure
- Decreases the function of visceral organs
- Increases the metabolic rate of skeletal muscles

Seeley et al., 2008

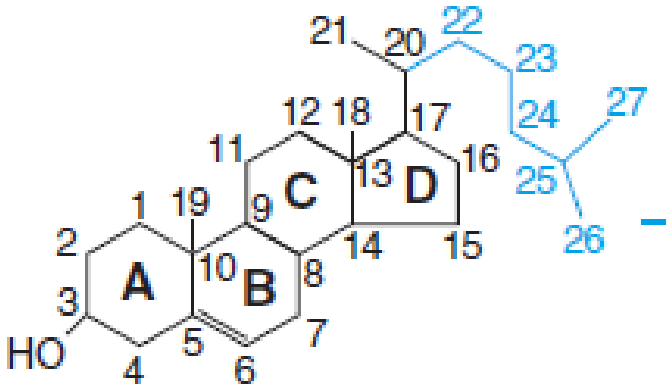
Pheochromocytoma: a tumor of chromaffin tissue, produces excessive catecholamines

Adrenal cortex

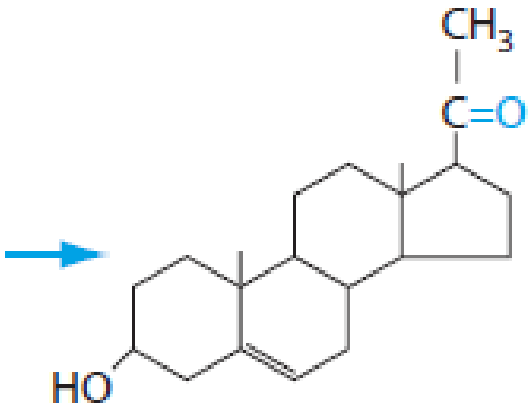


Zona glomerulosa consists of small clumps of cells.
Zona fasciculata consists of cords of cells oriented radially to the center of the gland.
Zona reticularis comprises interlaced strands of cells.

Adrenal steroid hormones

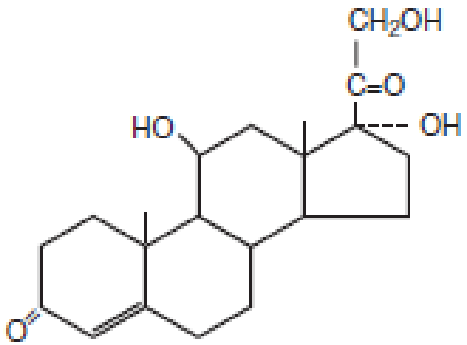


cholesterol

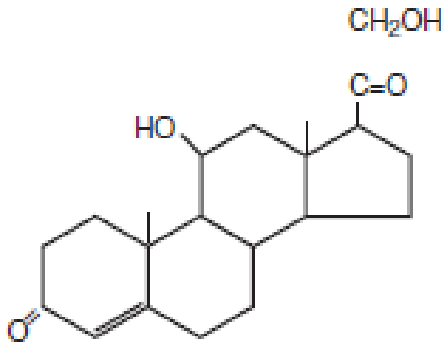


pregnenolone

Glucocorticoids

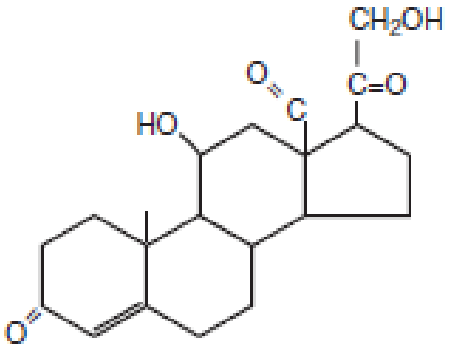


cortisol

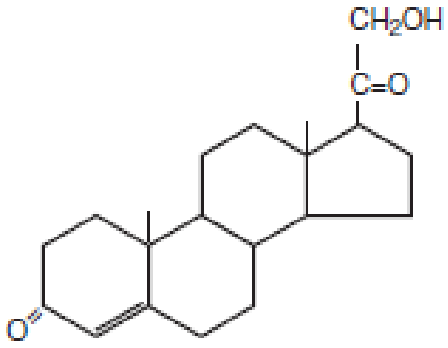


corticosterone

Mineralocorticoids

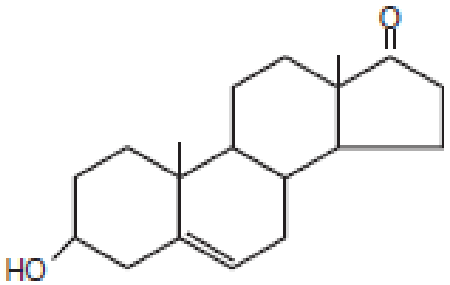


aldosterone

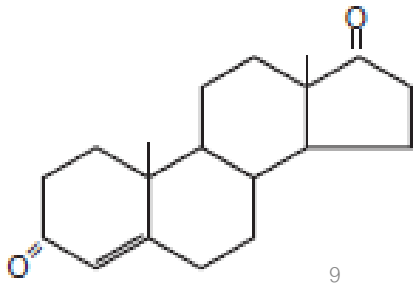


11-deoxycorticosterone (DOC)

Androgens

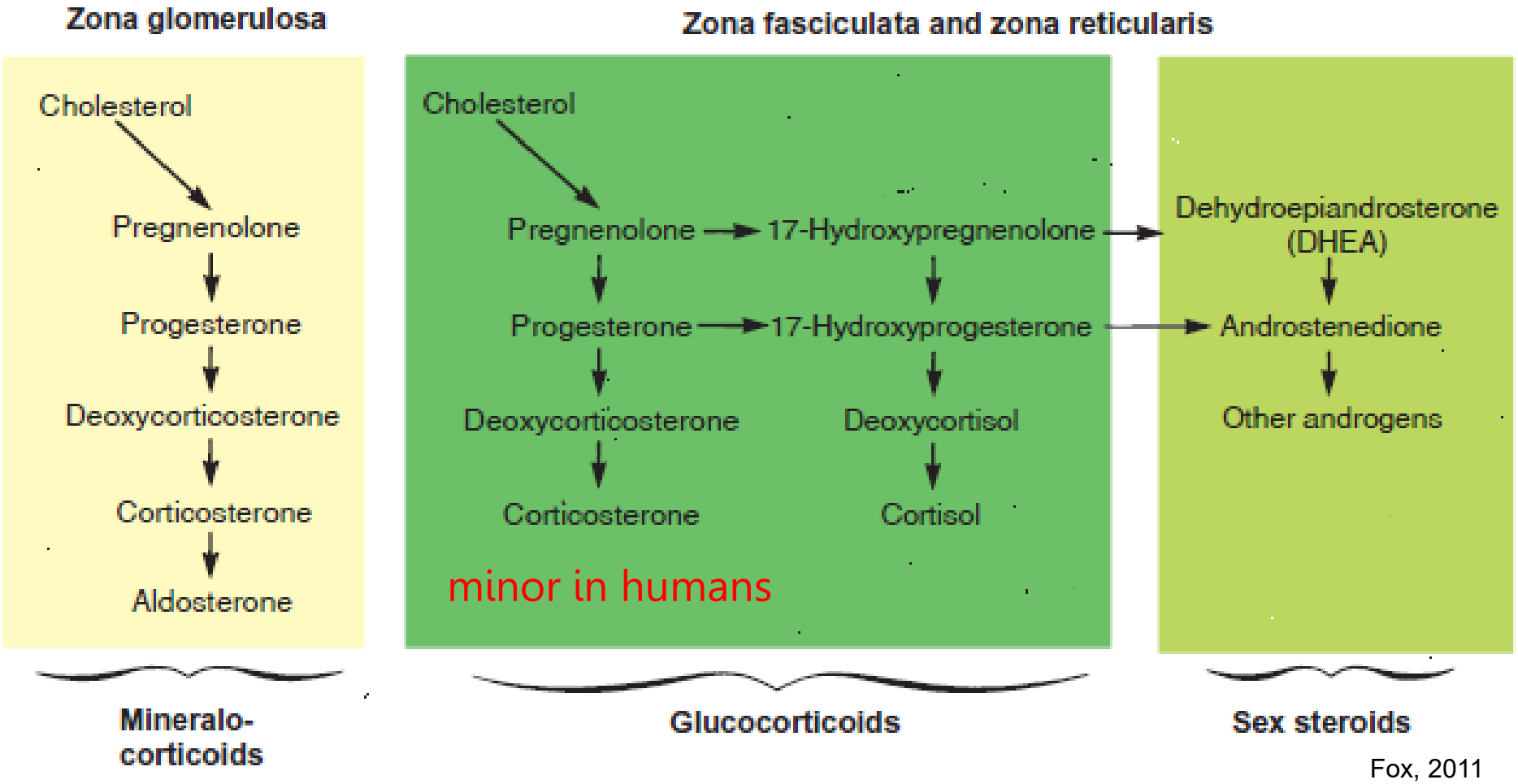


dehydroepiandrosterone



androstenedione

Johnson, 2003

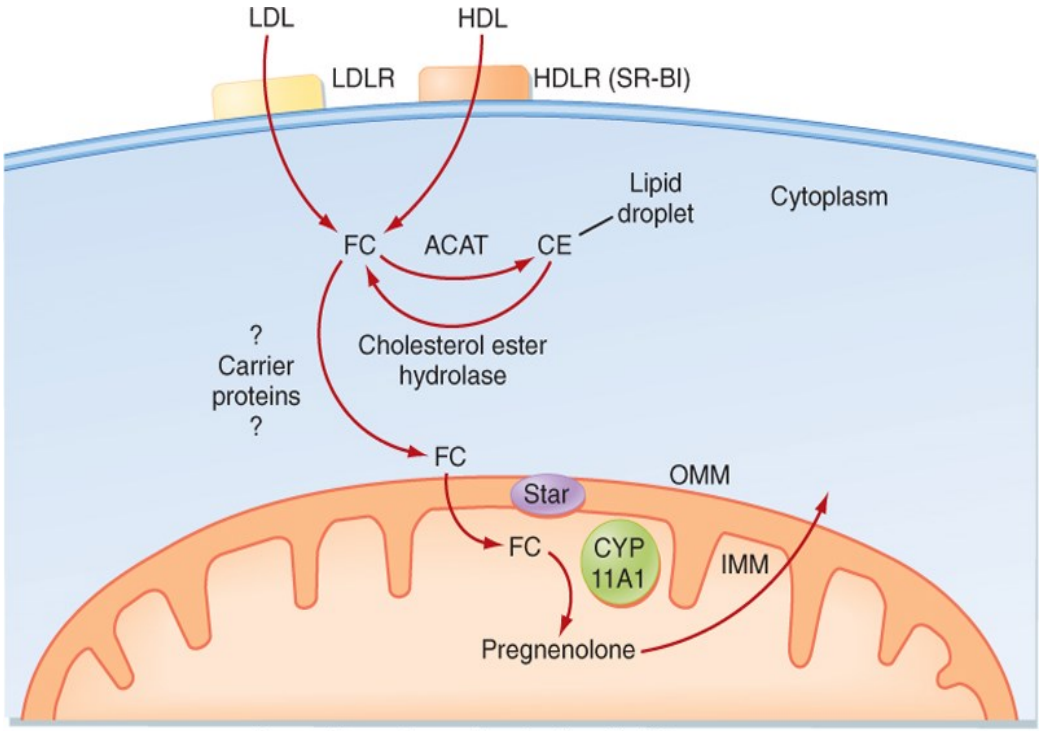


Fox, 2011

Deoxycorticosterone (DOC): a weak mineralocorticoid, ⇧ levels of DOC → hypertension

Zona fasciculata

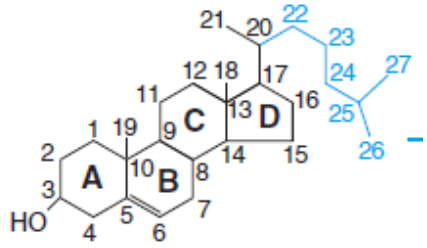
- Foamy cytoplasm (filled with lipid droplets)
- Make cholesterol de novo & import from blood in LDL and HDL (minimal in humans) forms.
- FC is esterified & stored in lipid droplets and turned back into FC by cholesterol ester hydrolase.
- Produces glucocorticoid hormone: **cortisol**



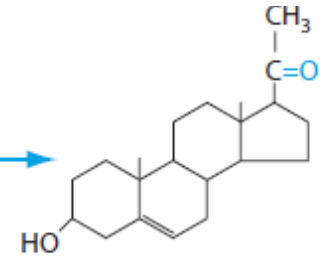
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LDLR, low-density lipoprotein receptor
HDLR, high-density lipoprotein receptor
FC, free cholesterol
ACAT, acyl CoA: cholesterol acyltransferase
CE, cholesterol esters
StAR, steroidogenic acute regulatory protein **(transporting FC to the inner mitochondrial membrane)**
IMM, inner mitochondrial membrane
OMM, outer mitochondrial membrane

Steroidogenic pathways



cholesterol



pregnenolone

(Koeppen and Stanton, 2008)

Zona Glomerulosa

Zona Fasciculata

Zona Reticularis

Cholesterol

CYP11A1 removal of side-chain

Mitochondria

Pregnenolone

3β-hydroxysteroid dehydrogenase (HSD)

Progesterone

CYP21A2 21-hydroxylase

sER

11-Deoxycorticosterone DOC

CYP11B2 11-hydroxylase

Mitochondria

Corticosterone

CYP11B2 18-hydroxylase

Mitochondria

18 (OH) Corticosterone

CYP11B2 18-oxidase

Mitochondria

Aldosterone

Cholesterol

CYP11A1 removal of side-chain

Mitochondria

Pregnenolone

Progesterone

17α-hydroxylase

3β-HSD

17α(OH)Progesterone

21β-hydroxylase

11-deoxycortisol

11β-hydroxylase

Cortisol

Cholesterol

CYP11A1 removal of side-chain

Mitochondria

Pregnenolone

17α-hydroxylase

17α(OH)Pregnenolone

CYP17 ,20-lyase

sER

DHEA

SULT2A1 sulfo-transferase

sER

DHEAS

17,20-lyase

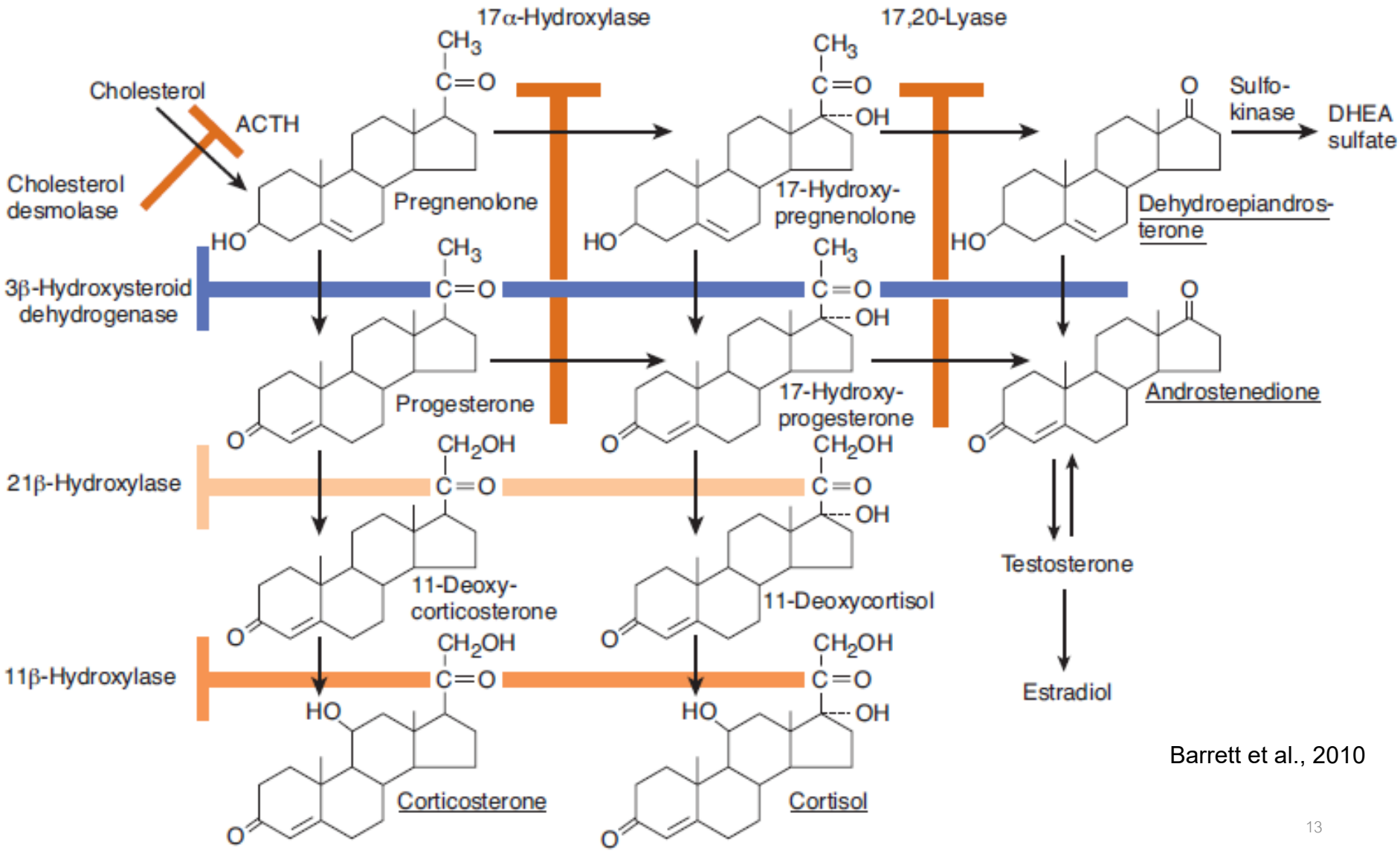
DHEA

3β-HSD conversion of Δ5 to Δ4 steroid

sER

Androstenedione (minor product)

Hormone biosynthesis in the zona fasciculata and zona reticularis



Barrett et al., 2010

In the circulation

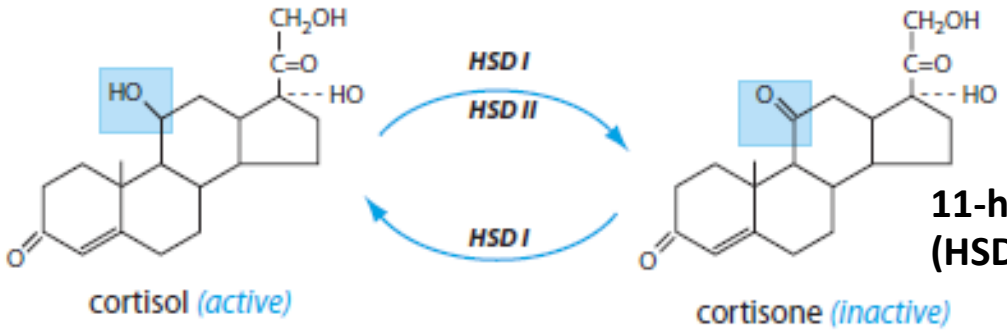
- 90% bind to corticosteroid-binding globulin (CBG)/transcortin
- 5–7% bind to albumin
- Circulating half-life is ~70 min

Liver

- Inactivate cortisol
- Conjugates active & inactive steroids with glucuronide or sulfate so that they can be excreted more readily by the kidney.

Tissue expressing glucocorticoid receptor: liver, adipose tissue, CNS, skin

Skin: cortisone-based creams can be applied to stop inflammation



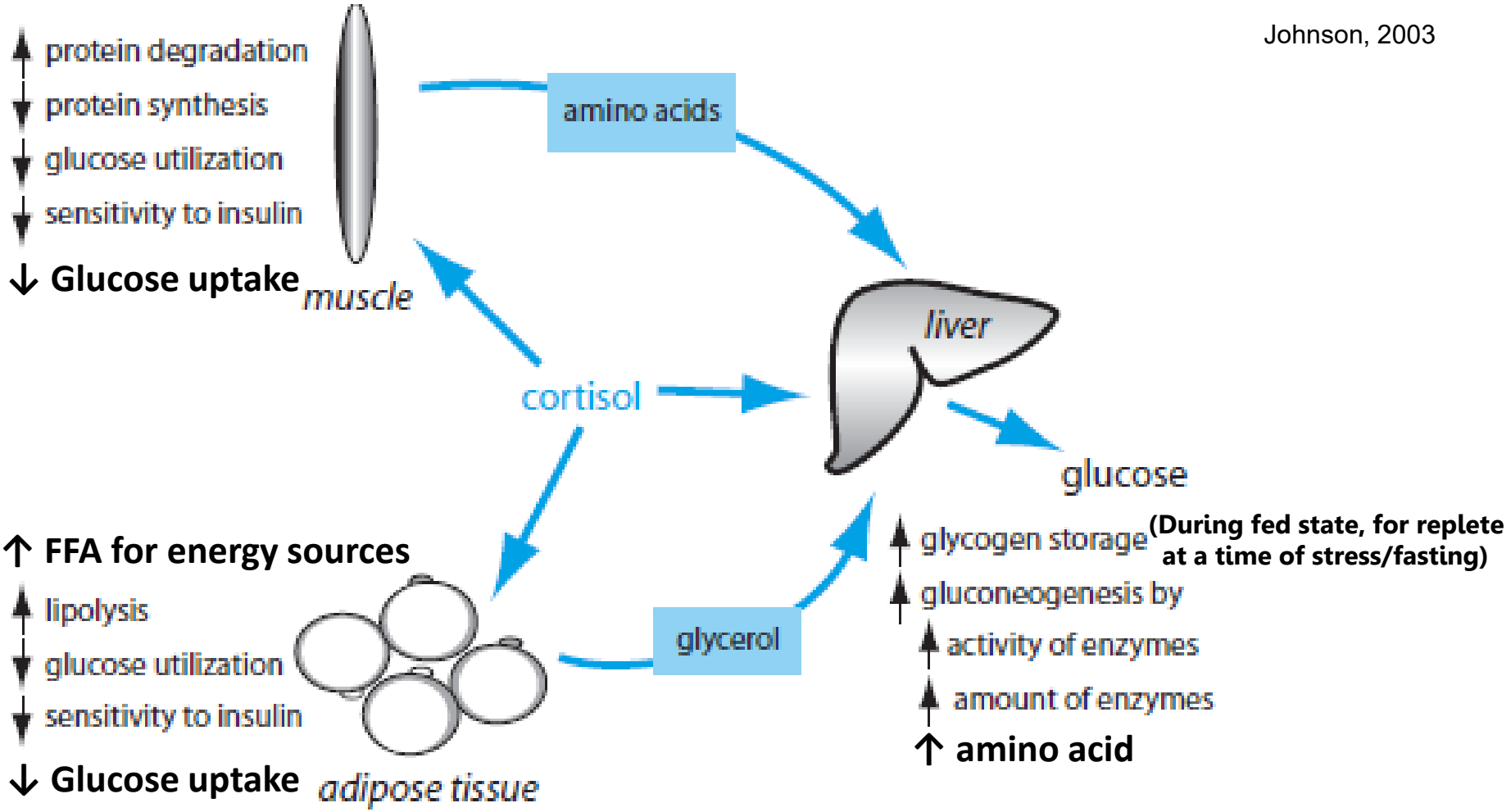
11-hydroxysteroid dehydrogenase (HSD)

Physiological actions of cortisol

- **A stress hormone** (physical and emotional)
- **Metabolic actions**: maintain blood glucose levels, CNS & CVS functions during fasting or stress
- **Permissive actions**: permit the catecholamines to induce vasoconstriction
- **Anti-inflammatory action**: interferes with almost every step of inflammation
- **Immunosuppressive action**: interfere with WBC
- ⇩ function of the reproductive axis at the hypothalamic, pituitary & gonadal levels
- ⇩ intestinal Ca^{2+} absorption & renal Ca^{2+} reabsorption, ⇧ bone resorption
- ⇩ fibroblast proliferation & collagen formation
- ⇩ secretion & action of ADH

- **Metabolic actions**: regulates blood glucose

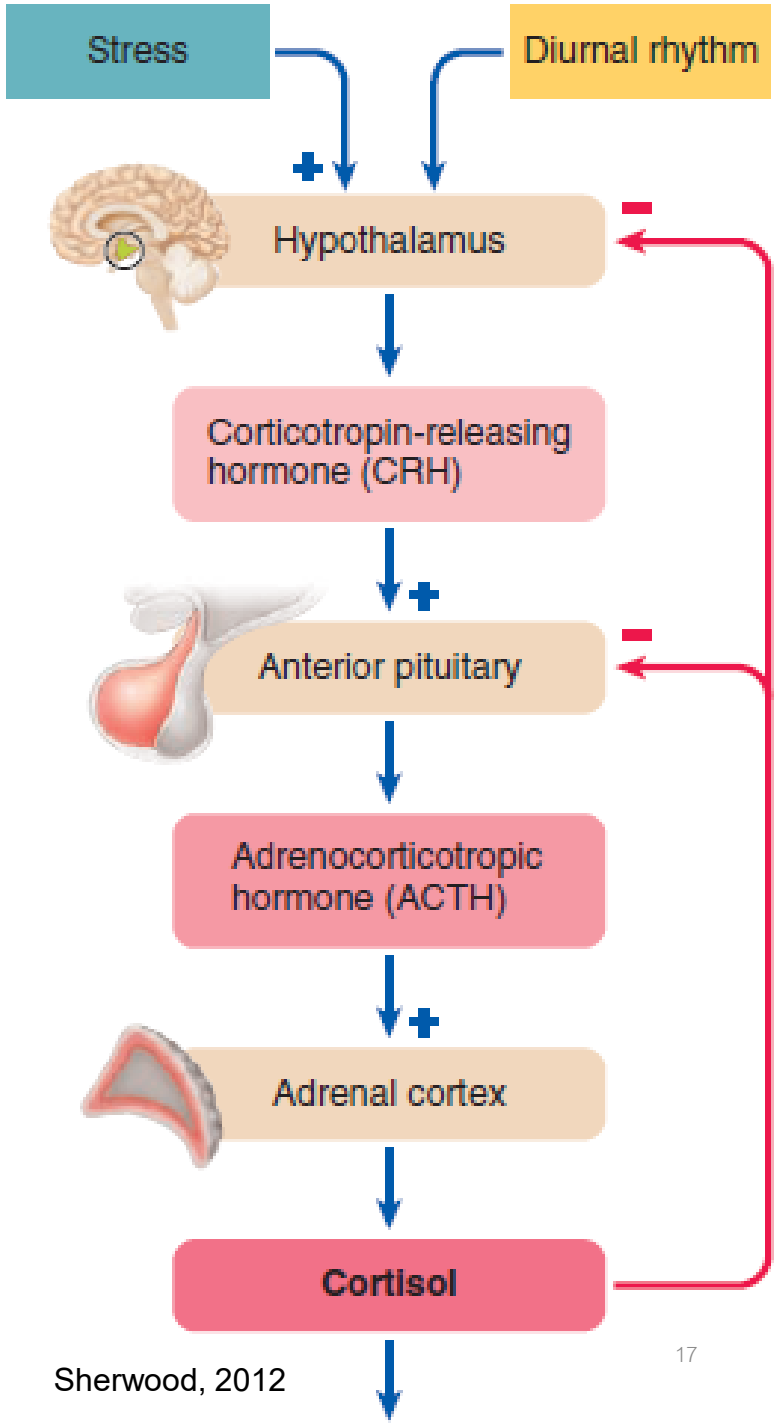
Johnson, 2003



Region specific: synergizes with insulin (during fed state/chronically ⇧ cortisol) to promote pre-adipocyte differentiation into adipocytes → ⇧ abdominal & interscapular adipose tissue

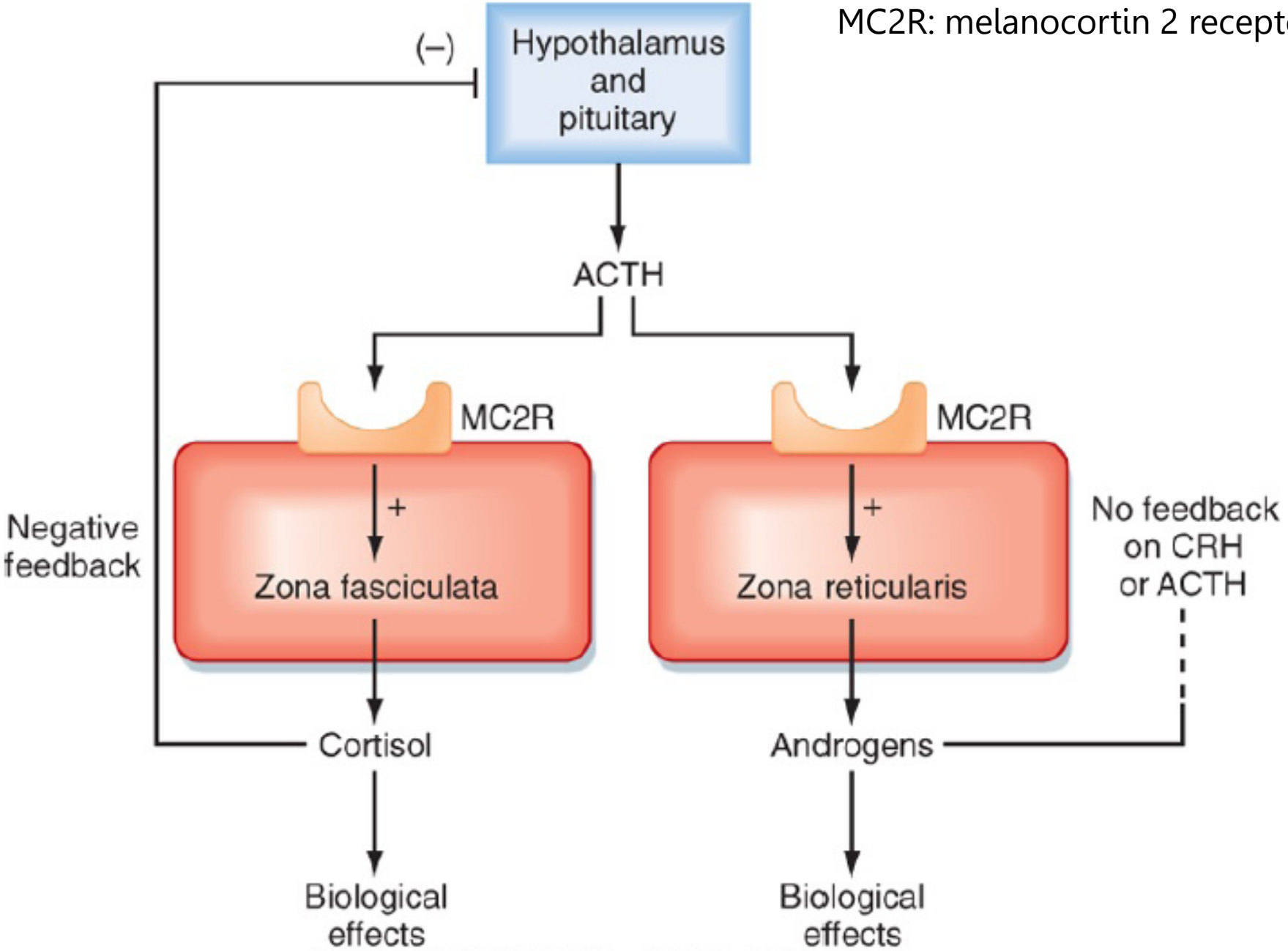
Regulation of cortisol production

- Hypothalamus-pituitary adrenal axis
- Neurogenic (e.g., fear) & systemic (e.g., hypoglycemia, hemorrhage, cytokines) forms of stress stimulate release of CRH



Sherwood, 2012

MC2R: melanocortin 2 receptor



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Cushing's syndrome

The signs and symptoms associated with prolonged exposure to inappropriately elevated glucocorticoids.

Causes

- Anterior pituitary tumour (Cushing's disease): ACTH-dependent conditions
- Adrenocortical tumour:

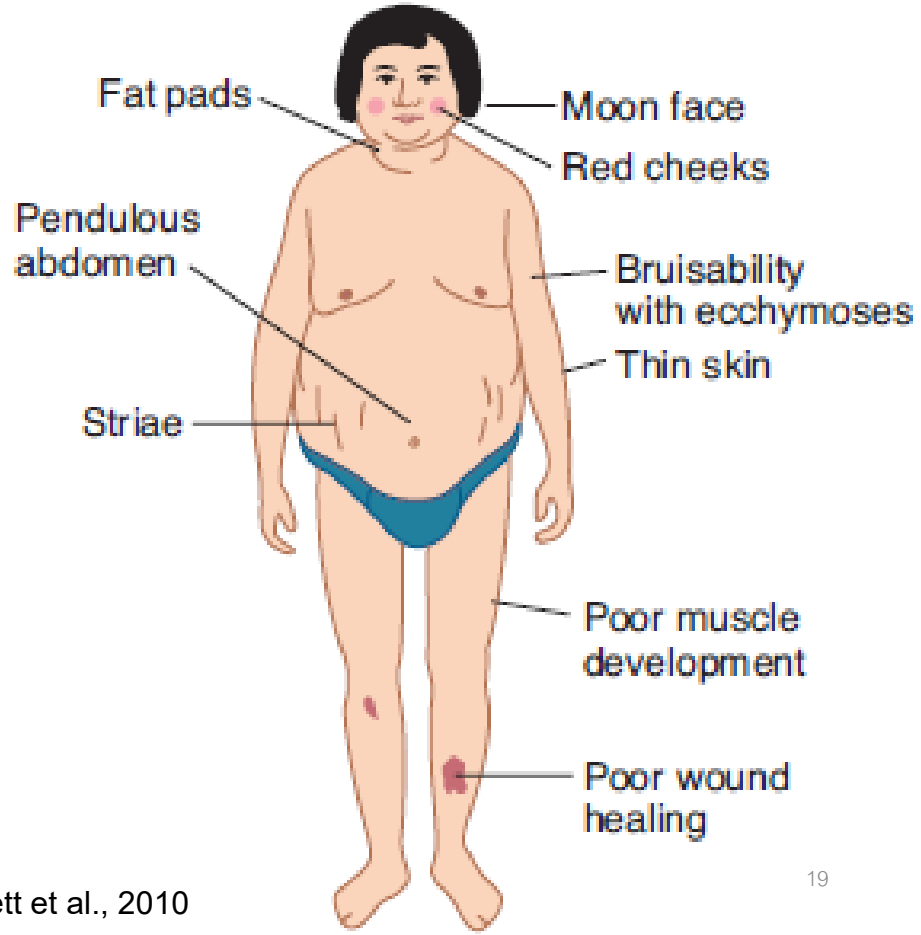
baseline ACTH suppressed

- Exogenous glucocorticoids:

baseline ACTH suppressed



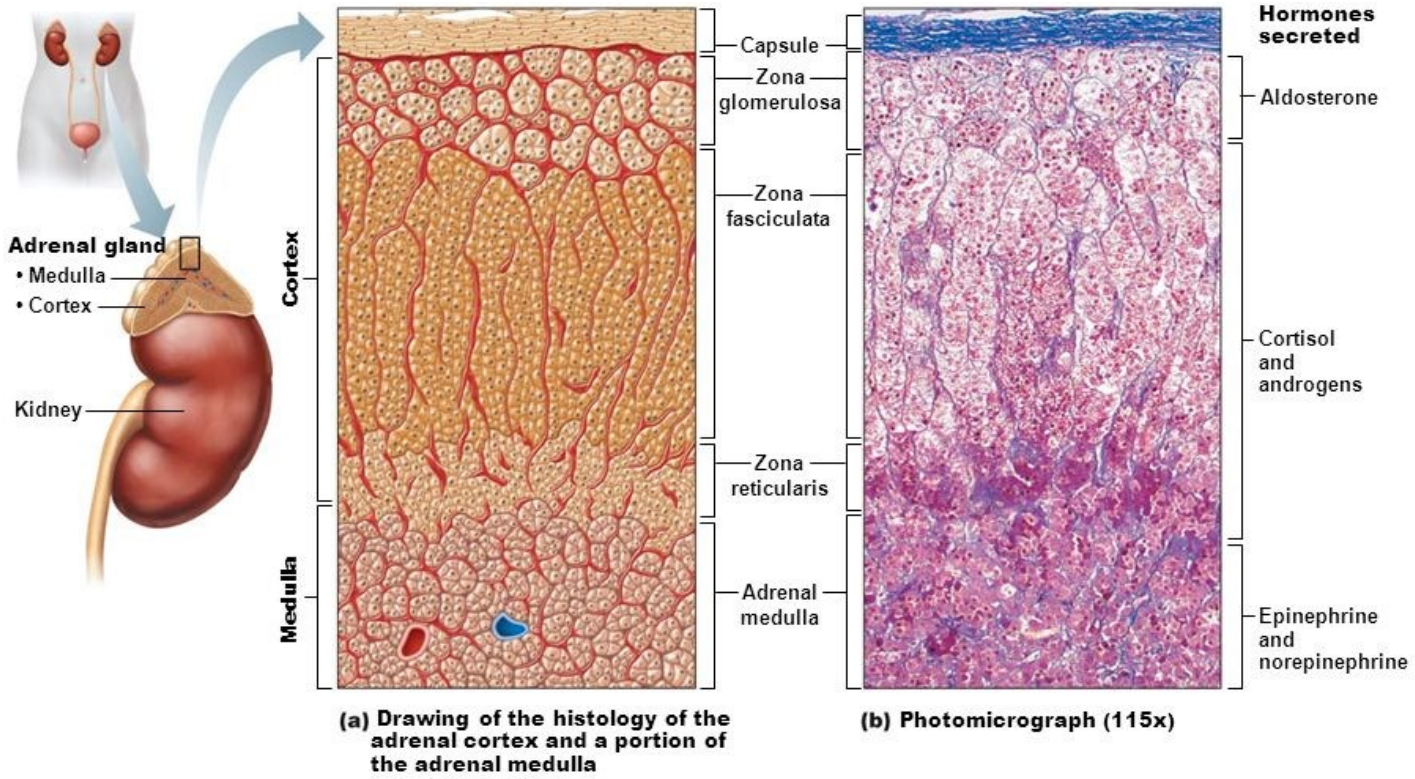
Holt and Hanley, 2012



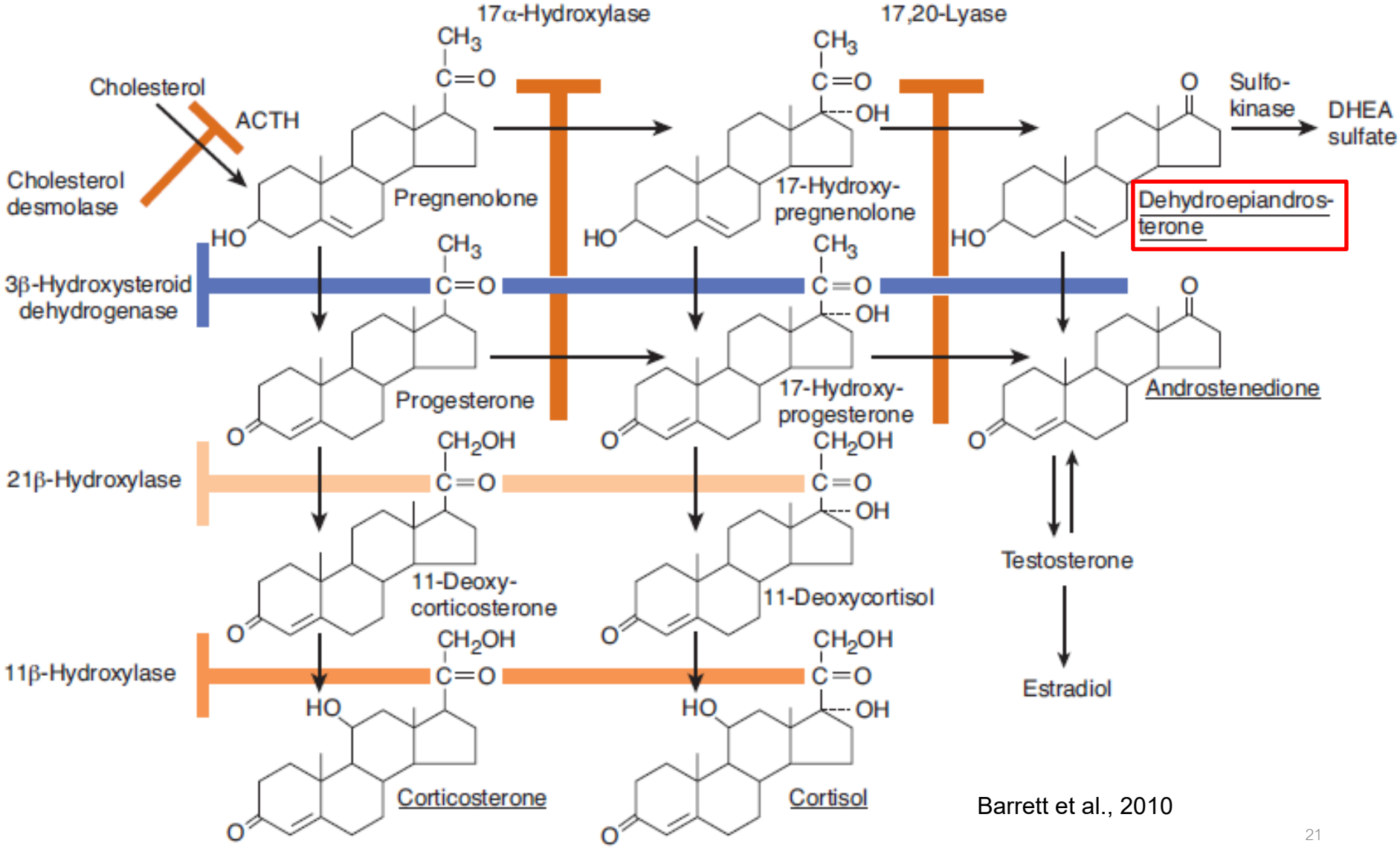
Barrett et al., 2010

Zona reticularis

- Appear after birth at about 5 years
- Produces adrenal androgens, especially **DHEAS** **(dehydroepiandrosterone sulfate)**, this onset is called adrenarche

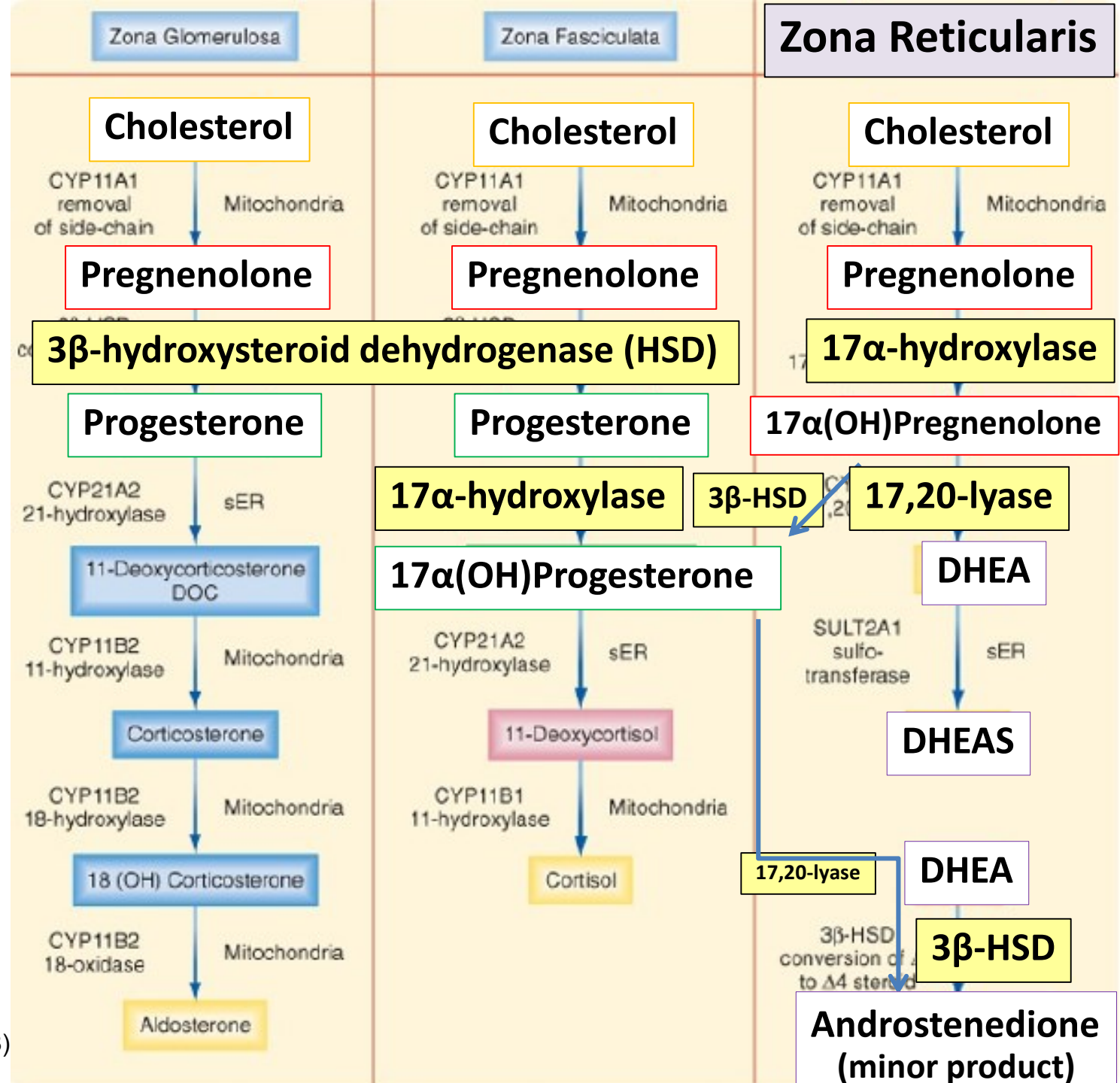


Hormone biosynthesis in the zona fasciculata and zona reticularis



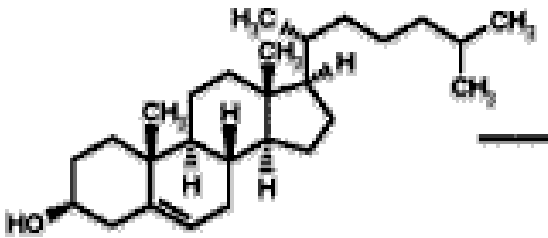
Barrett et al., 2010

Androgen synthesis



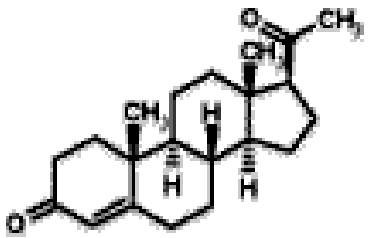
(Koeppen and Stanton, 2008)

Metabolism of DHEA and DHEAS



Cholesterol

Pregnenolone



Progesterone

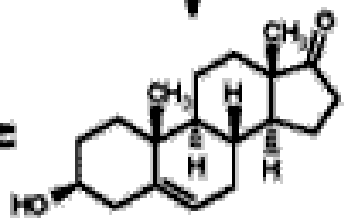
Cortisol

Aldosterone

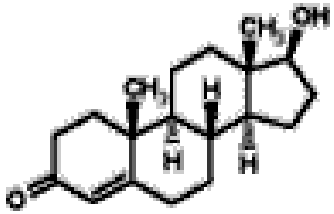


DHEAS

Sulfatase



DHEA



Testosterone and other androgens

Active androgen

- DHEA binds to albumin & other globulins in blood with low affinity, it is excreted efficiently by the kidney. (half-life: 15–30 min)
- DHEAS binds to albumin with very high affinity. (half-life: 7–10 h)

Physiological actions of adrenal androgens

Active androgen: (testosterone & dihydrotestosterone)

- In men: negligible
- In women: required for the growth of axillary & pubic hair, and for libido

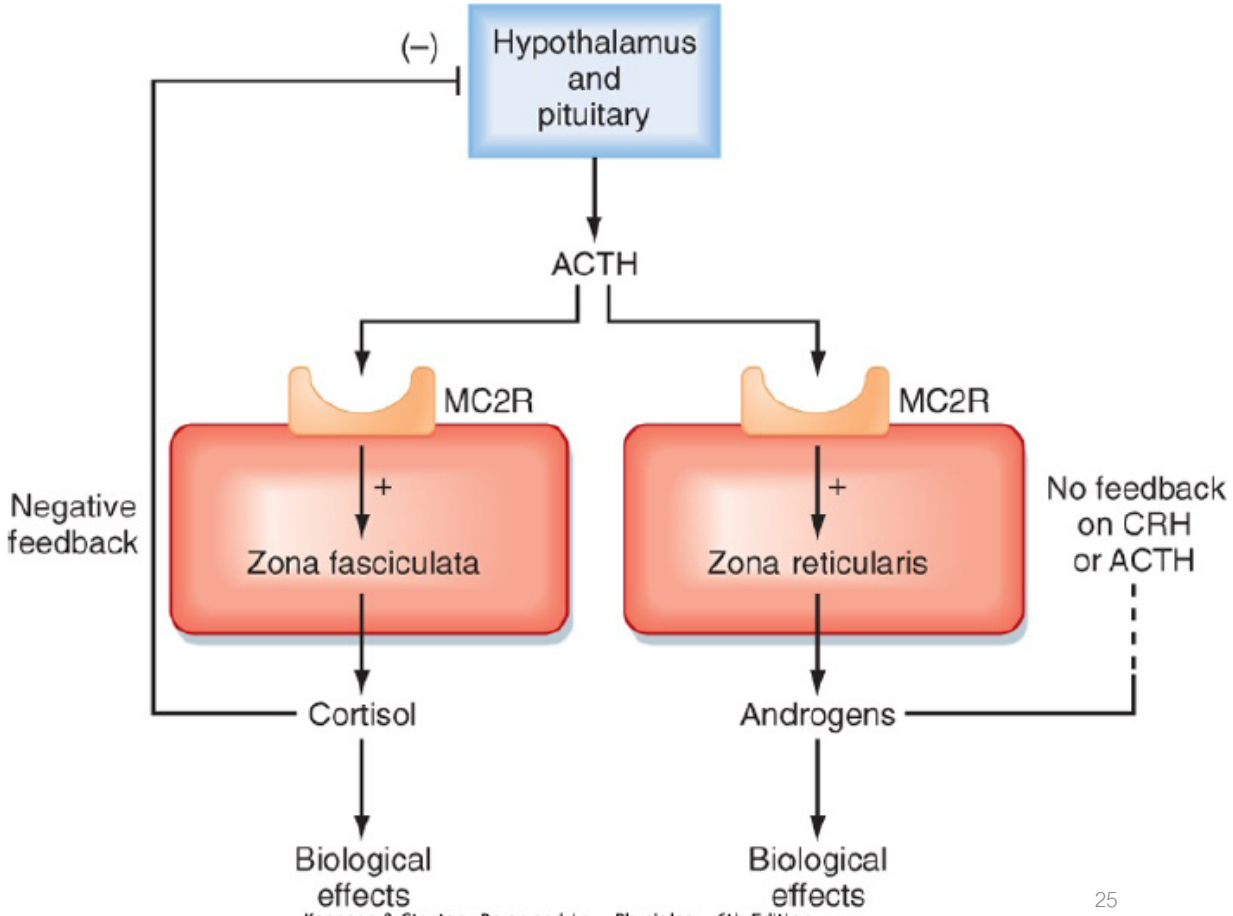
Adrenal androgen excess

- Adrenal tumor, Cushing's syndrome, congenital adrenal hyperplasia
- Masculinization of women
- Hirsutism: excessive facial and body hair



Regulation of zona reticularis function

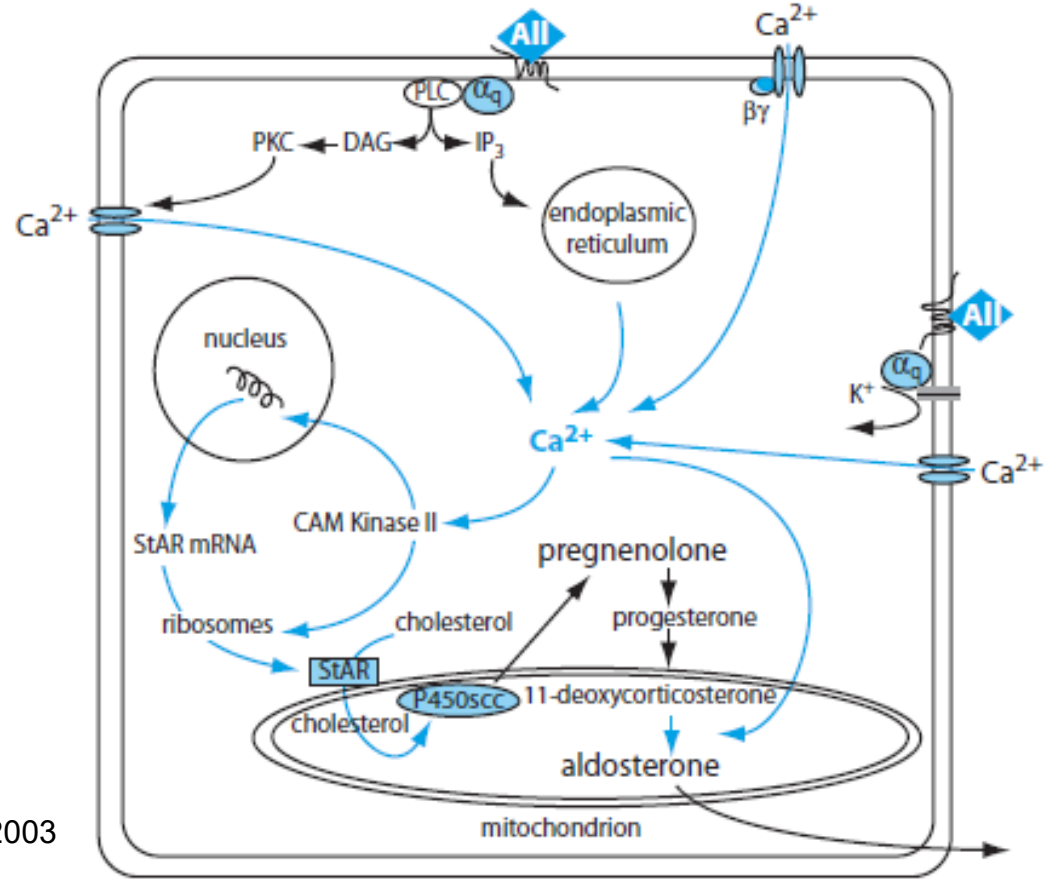
- Loophole in the hypothalamus-pituitary-adrenal axis
- If cortisol production is blocked (i.e., CYP11B1 deficiency), ACTH levels increase, along with adrenal androgen.



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Zona glomerulosa

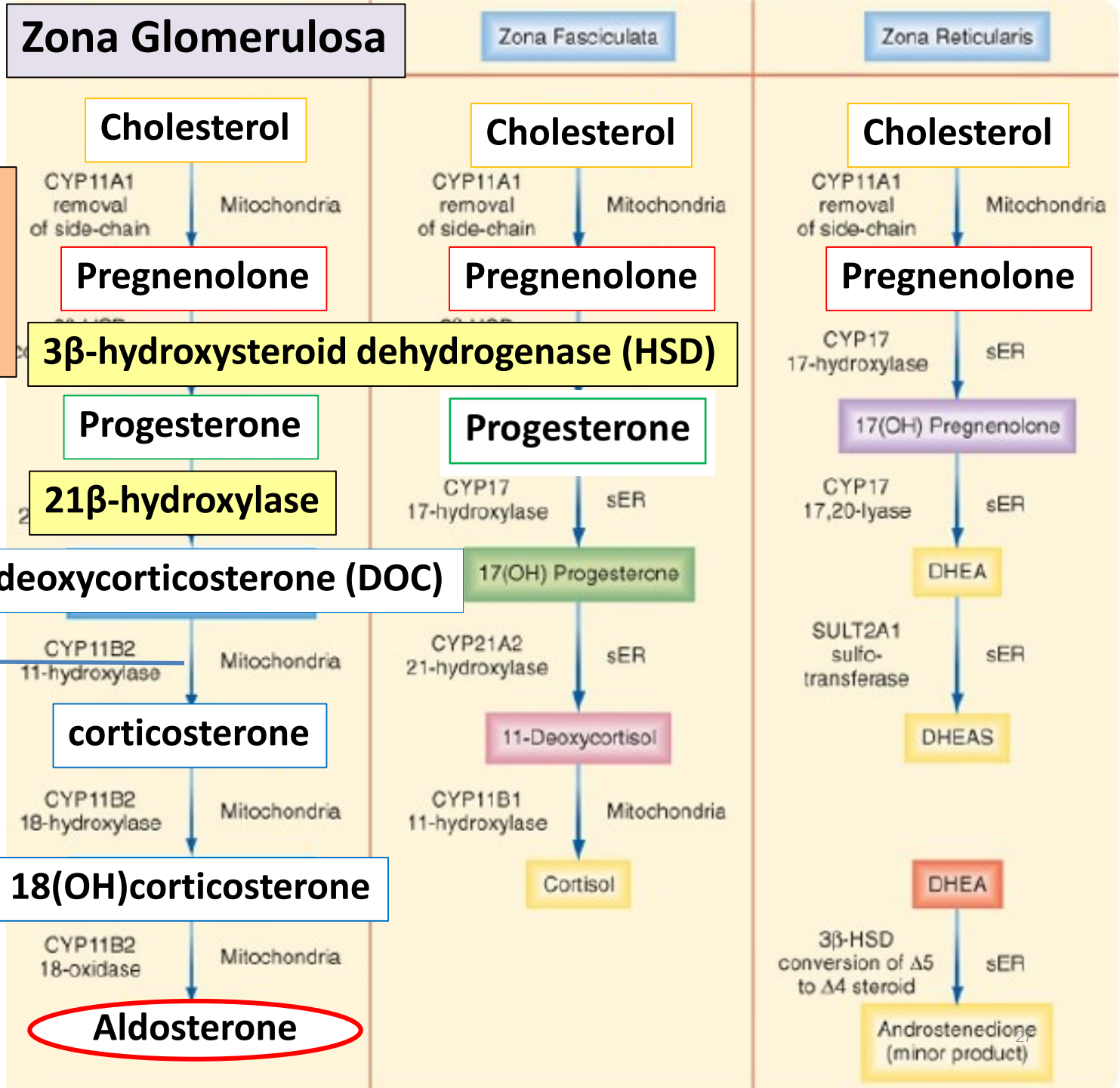
- Produces the mineralocorticoid **aldosterone**
- Minimally influenced by ACTH, but rather regulated by renin-angiotensin system, plasma [K^+], and atrial natriuretic peptide (ANP)
- Not express CYP17, so zona glomerulosa cells never make cortisol



Johnson, 2003

Steroidogenic pathways

Lacks 17α-hydroxylase: never make cortisol/ androgen, but contain aldosterone synthase



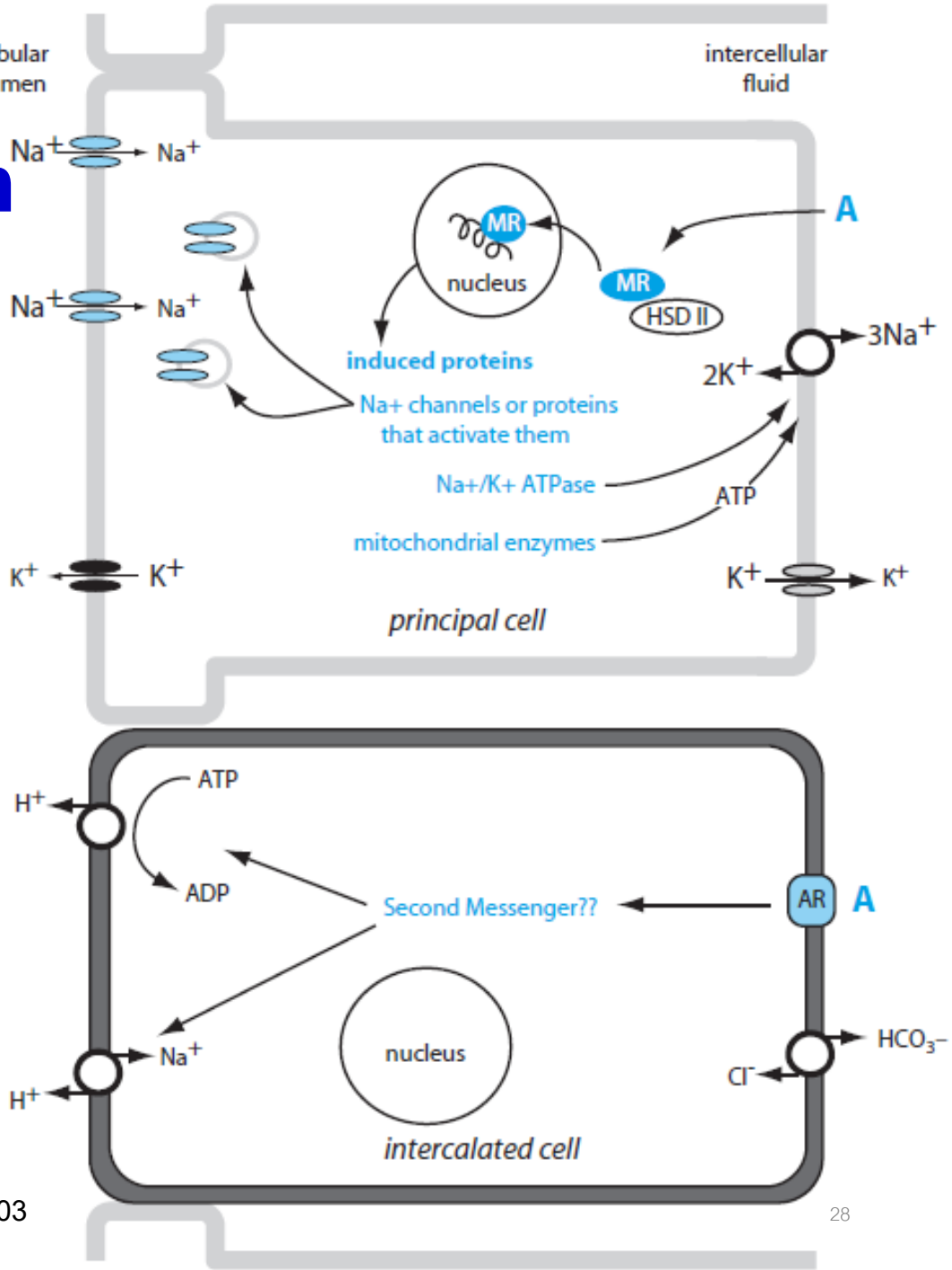
Mechanism of aldosterone action

- **Principal cells** of the cortical collecting ducts: aldosterone (A) → ⇧ transcription & translation of new proteins.

MR: mineralocorticoid receptor

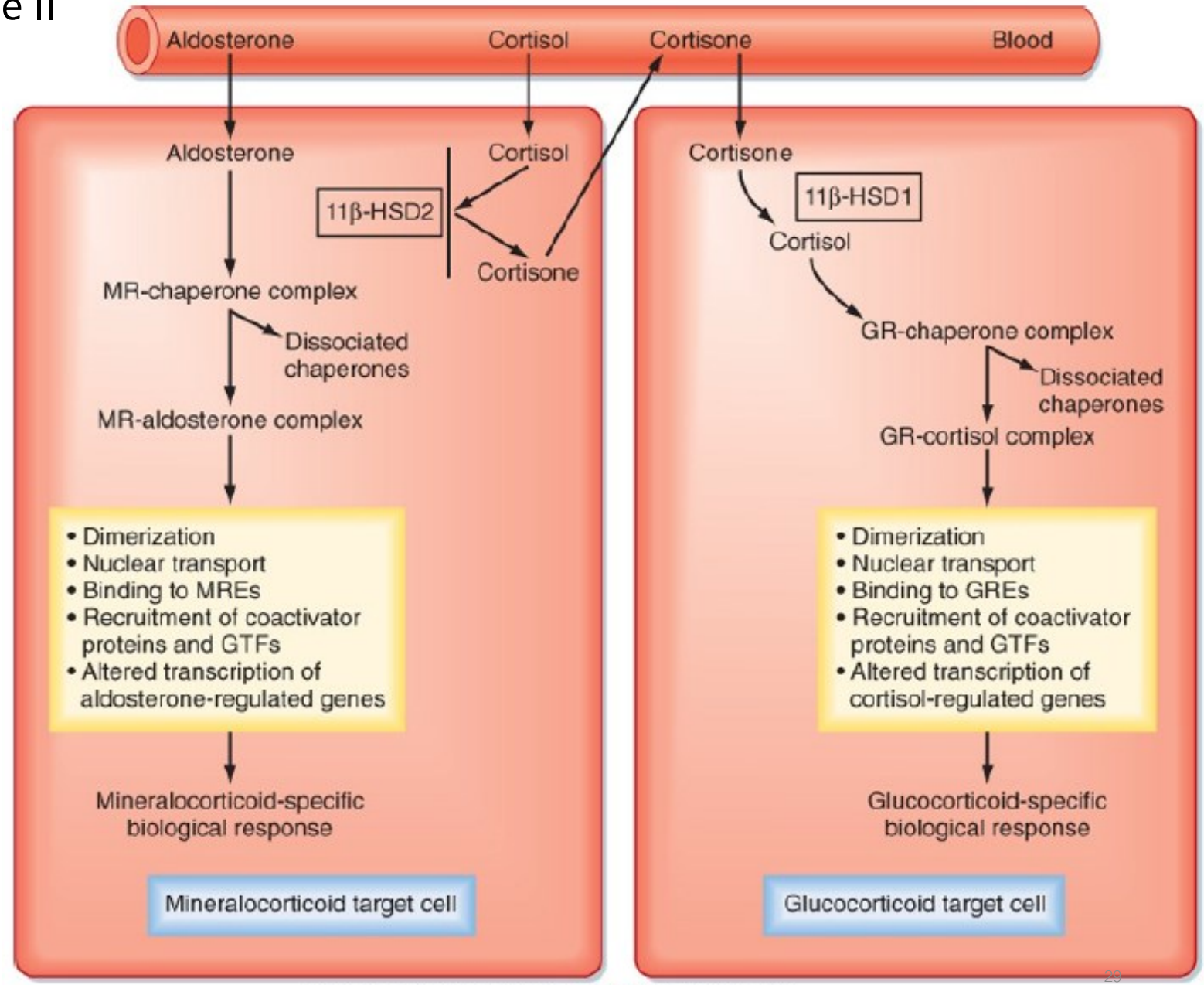
HSD-II: 11-hydroxysteroid dehydrogenase II

- **Intercalated cells**: A → ⇧ H^+ secretion (probably involves receptor on the cell surface, (AR)

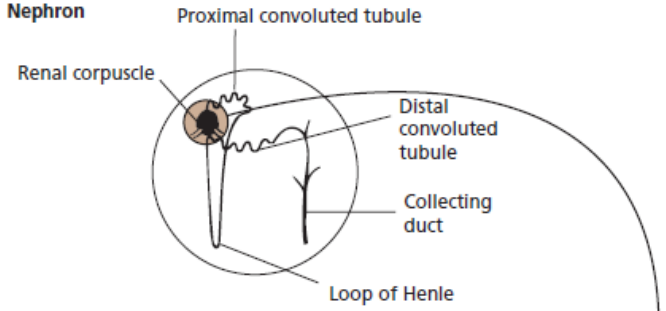


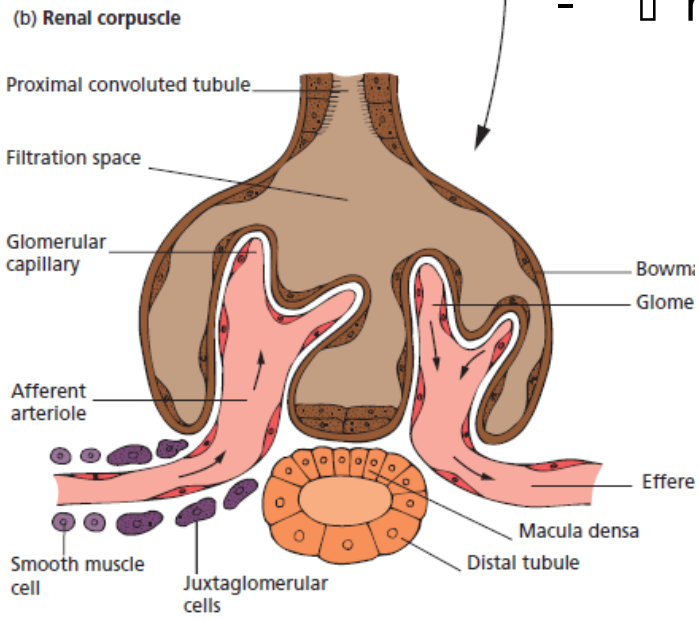
Johnson, 2003

MR is protected from activation by cortisol by 11β-hydroxysteroid dehydrogenase II



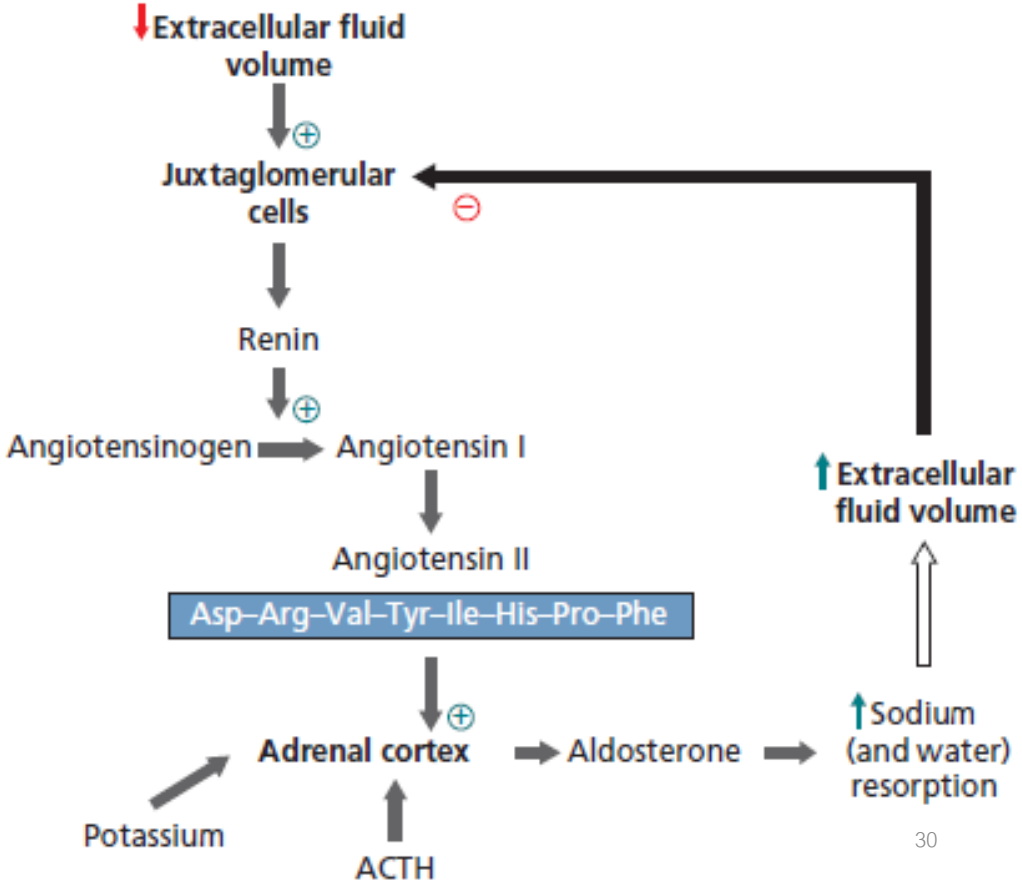
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secrete renin by

- ⇩ BP in the afferent arterioles of the glomeruli
- ⇩ [NaCl] in renal tubular fluid at the macula densa
- ⇧ renal sympathetic nerve activity



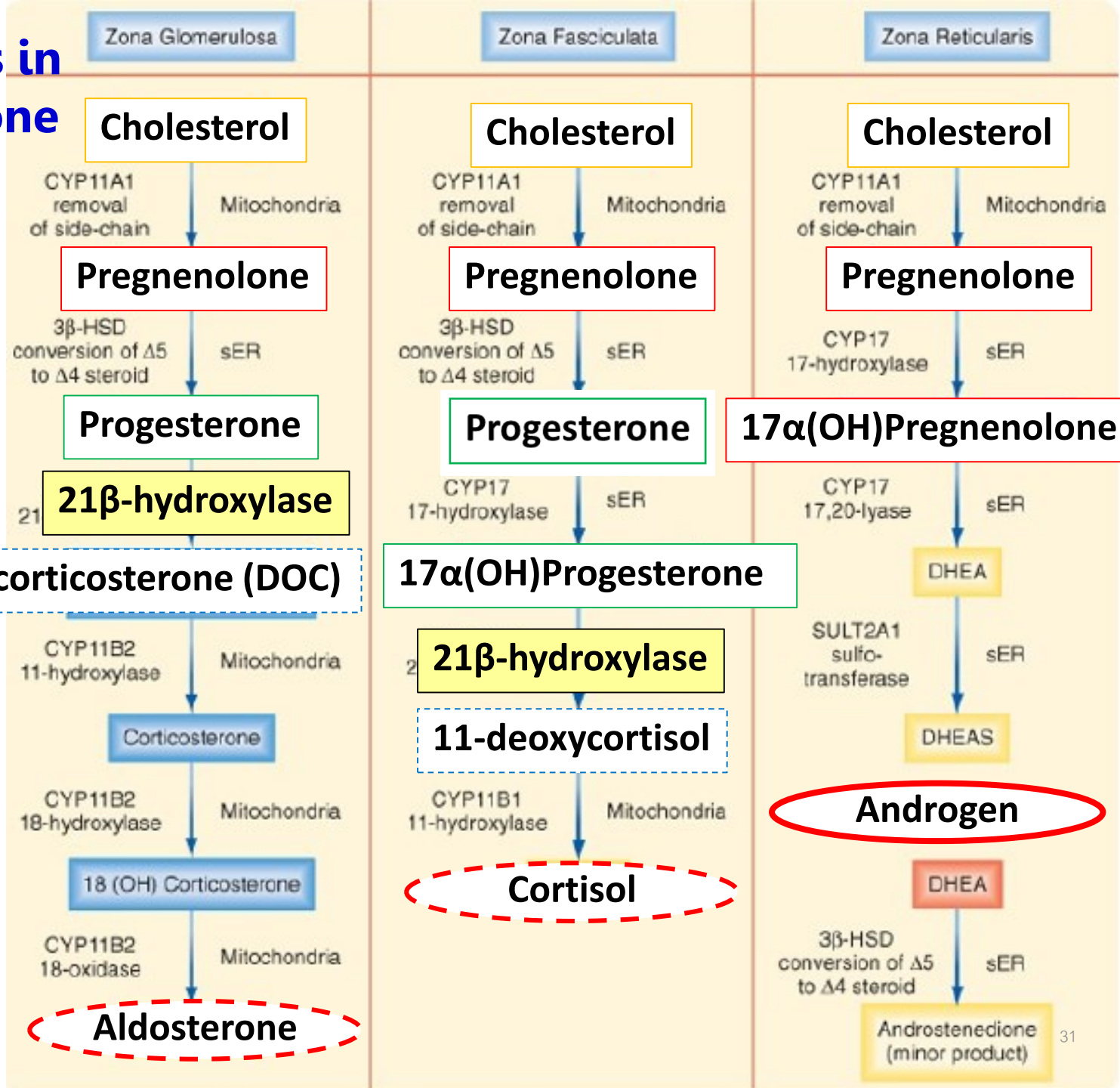
Holt and Hanley, 2012

Abnormalities in steroid hormone biosynthesis

Congenital Adrenal Hyperplasia from ⇧ ACTH

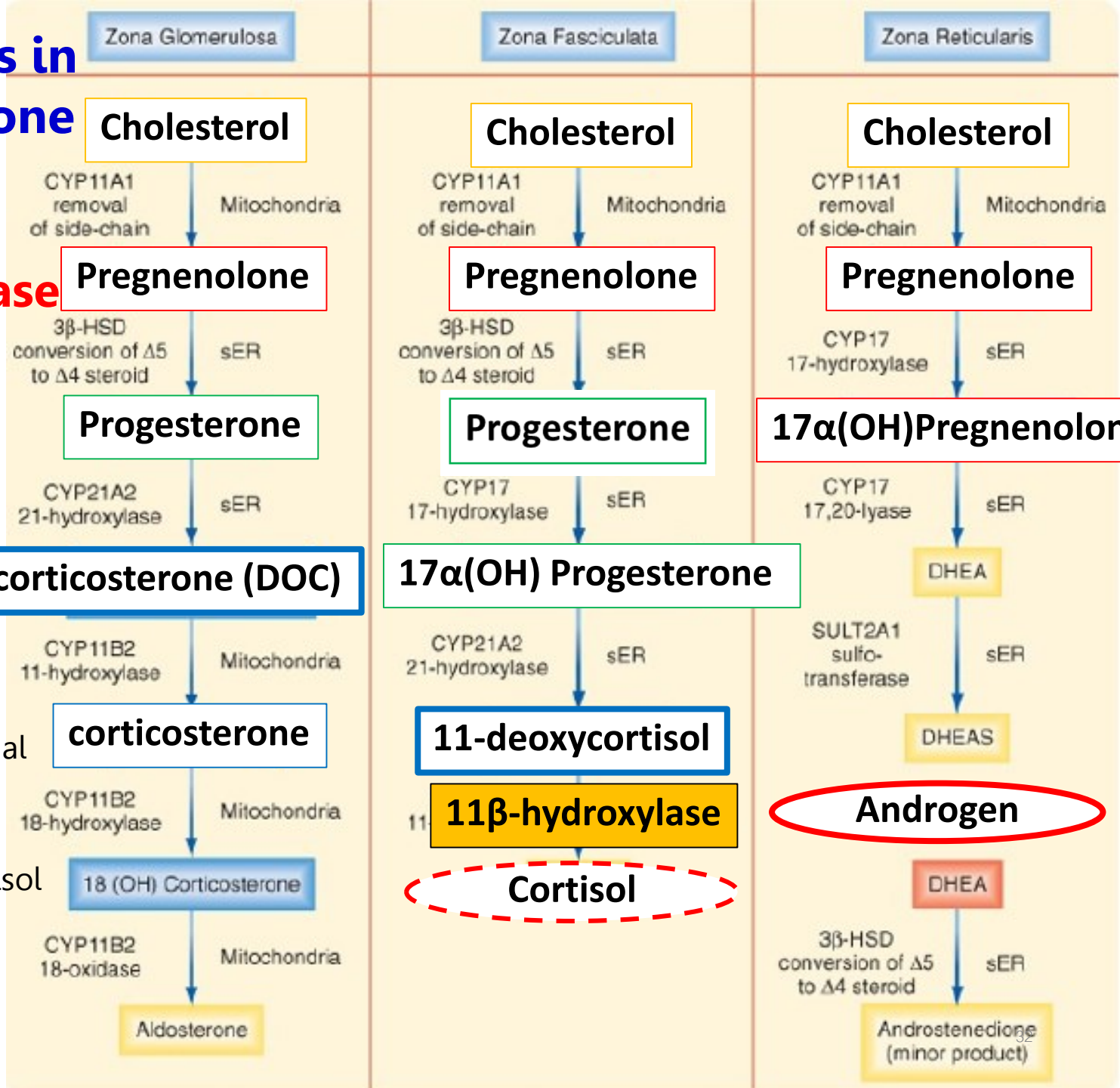
21β-hydroxylase deficiency

95% of genetic abnormalities in adrenal steroid hormone synthesis



Abnormalities in steroid hormone biosynthesis

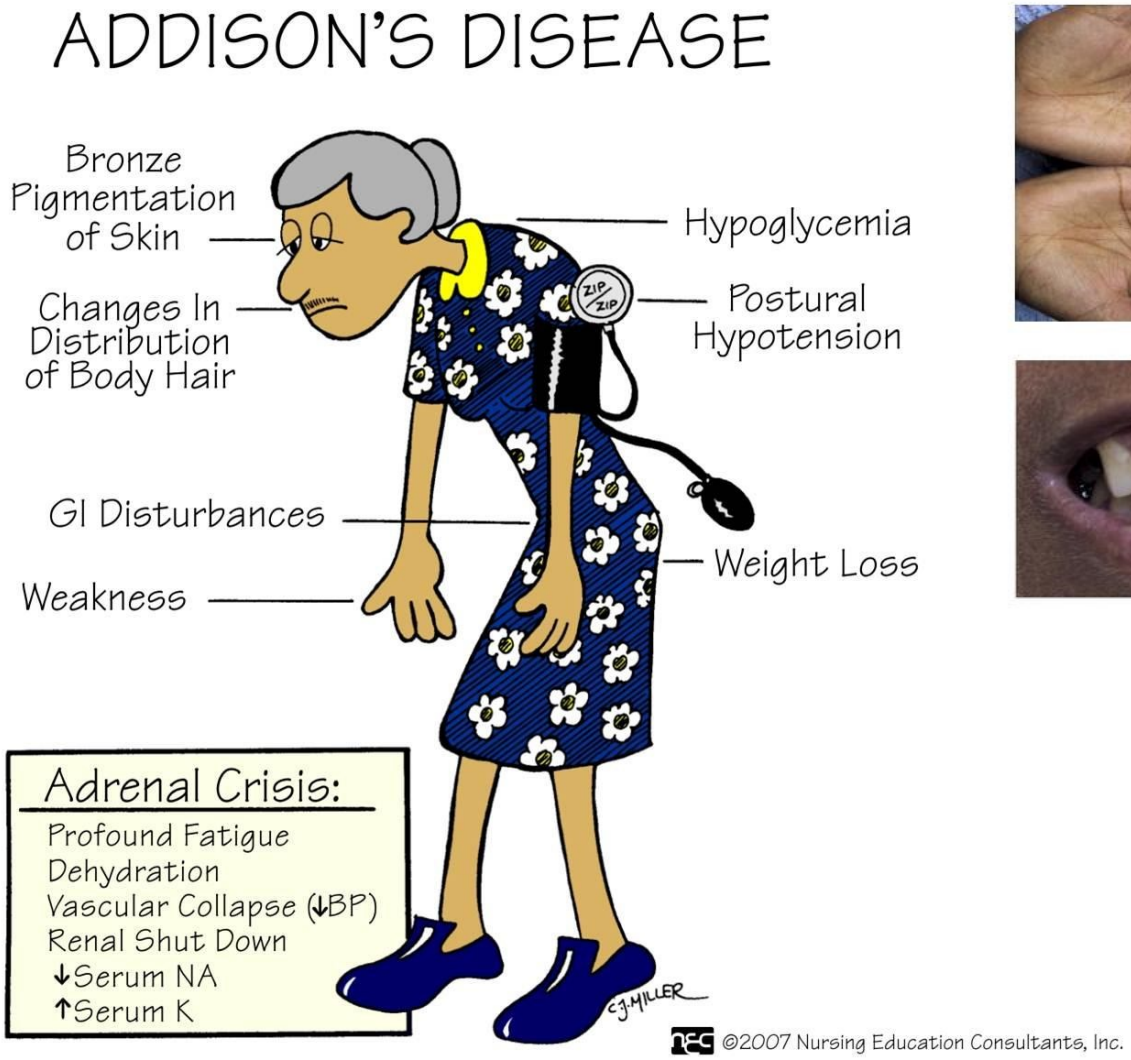
11β-hydroxylase deficiency



- 2nd most frequent abnormalities in adrenal steroid hormone synthesis
- Excess 11-deoxycortisol & 11-DOC (active mineralocorticoid activity): retain salt & water

Adrenal Insufficiency

- **Adrenal insufficiency may be caused by**
  - Destruction of the adrenal cortex (primary adrenal insufficiency)
  - ⇩ ACTH secretion (secondary adrenal insufficiency)
  - ⇩ CRH secretion (tertiary adrenal insufficiency)
- **Addison disease (primary adrenal insufficiency)**
  - Destruction of the adrenal gland by microorganisms or autoimmune disease
  - All 3 zones are usually involved.
  - Inadequate secretion of glucocorticoids, mineralocorticoids & androgens







Addison's disease:



- Note the generalised skin pigmentation (in a Caucasion patient) but especially the deposition in the palmer skin creases, nails and gums.

- She was treated many years ago for pulmonary TB. What are the other causes of this condition?

- ⇧ ACTH secretion, binds MC1R in melanocytes (⇧ skin pigmentation).
- Treatment for acute adrenal insufficiency should be directed at the reversal of the hypotension and electrolyte abnormalities.

Table 18–5 The Adrenal Hormones

Region/Zone	Hormone	Primary Target	Hormonal Effect	Regulatory Control
CORTEX				
Zona glomerulosa	Mineralocorticoids (primarily aldosterone)	Kidneys	Increase renal reabsorption of Na^+ and water (especially in the presence of ADH) and accelerate urinary loss of K^+	Stimulated by angiotensin II, elevated plasma K^+ or a fall in plasma Na^+; inhibited by ANP and BNP
Zona fasciculata	Glucocorticoids (cortisol [hydrocortisone], corticosterone)	Most cells	Release of amino acids from skeletal muscles and lipids from adipose tissues; promote liver formation of glucose and glycogen; promote peripheral utilization of lipids; anti-inflammatory effects	Stimulated by ACTH from the anterior lobe of the pituitary gland
Zona reticularis	Androgens	Most cells	Not important in adult men; encourages bone growth, muscle growth, and blood formation in children and women	Stimulated by ACTH from the anterior lobe of the pituitary gland
MEDULLA				
	Epinephrine, norepinephrine	Most cells	Increases cardiac activity, blood pressure, glycogen breakdown, blood glucose levels; releases lipids by adipose tissue	Stimulated during sympathetic activation by sympathetic preganglionic fiber

Martini et al., 2012

TABLE 18.9 Symptoms of Hyposecretion and Hypersecretion of Adrenal Cortex Hormones

Hyposecretion	Hypersecretion
Mineralocorticoids (Aldosterone)	
Hyponatremia (low blood levels of sodium)	Slight hypernatremia (high blood levels of sodium)
Hyperkalemia (high blood levels of potassium)	Hypokalemia (low blood levels of potassium)
Acidosis	Alkalosis
Low blood pressure	High blood pressure
Tremors and tetany of skeletal muscles	Weakness of skeletal muscles
Polyuria	Acidic urine
Glucocorticoids (Cortisol)	
Hypoglycemia (low blood glucose levels)	Hyperglycemia (high blood glucose levels; adrenal diabetes)—leads to diabetes mellitus
Depressed immune system	Depressed immune system
Protein and fats from diet unused, resulting in weight loss	Destruction of tissue proteins, causing muscle atrophy and weakness osteoporosis, weak capillaries (easy bruising), thin skin, and impaired wound healing; mobilization and redistribution of fats, causing depletion of fat from limbs and deposition in face (moon face), neck (buffalo hump), and abdomen
Loss of appetite, nausea, and vomiting	Emotional effects, including euphoria and depression
Increased skin pigmentation (occurs if ACTH levels are elevated)	
Androgens	
In women, reduction of pubic and axillary hair	In women, hirsutism (excessive facial and body hair), acne, increased sex drive, regression of breast tissue, and loss of regular menses

Seeley et al., 2008


I <3 FLUFFY ANIMALS!
I SHOULD STUDY BIOLOGY!
Aldosterone is a hormone of the adrenal gland, which increases sodium reabsorption in the collecting ducts of the kidneys, whereas vasopressine, secreted by...
WHAT HAVE I DONE...